

CLAS12 Collaboration Meeting

ϕ electroproduction on the neutron with RGB

11 March 2026

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Summary

1. Motivations
2. Events selection
3. Generator for MC simulation
4. Effect of different cuts
5. Yield extraction
6. Neutron Efficiency correction
7. Acceptance correction
8. Binning correction
9. Preliminary plot of differential cross section

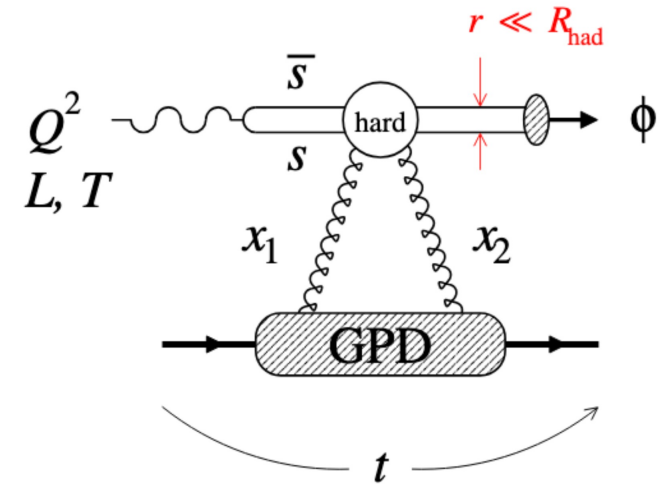
Introduction

Analysis objective :

- Measurement of the cross section and differential cross section of the electroproduction of ϕ in the $K^+ K^-$ channel with **RGB** data (**deuterium** target).

$e n \rightarrow e n' K^+ K^- \rightarrow$ Access to the gluon GPD of the neutron

- We expect that cross section measurement to be similar to that of proton. We will compare with proton bound target (RGB) and with the work on free proton target (Bhawani's work on RGA).



Change of t -dependence with $Q^2 \rightarrow$
gluonic radius of the neutron

Events selection

Analysis :

$e n \rightarrow e n' K^+ K^-$

Datas :

sidisdvcs **RG-B outbending fall 2019**

Events selection :

e^-	K^+	K^-	neutron	proton
1	1	1	≥ 1	0

(select the best neutron which minimize the angle between the **missing nucleon** and the **neutron**)

$Q^2 > 1.0 \text{ GeV}^2$	$P_{\text{electron}} > 2 \text{ GeV}$	$\theta_{\text{neutron}} > 4^\circ$	$P_{\text{neutron}} > 0.25 \text{ GeV}$
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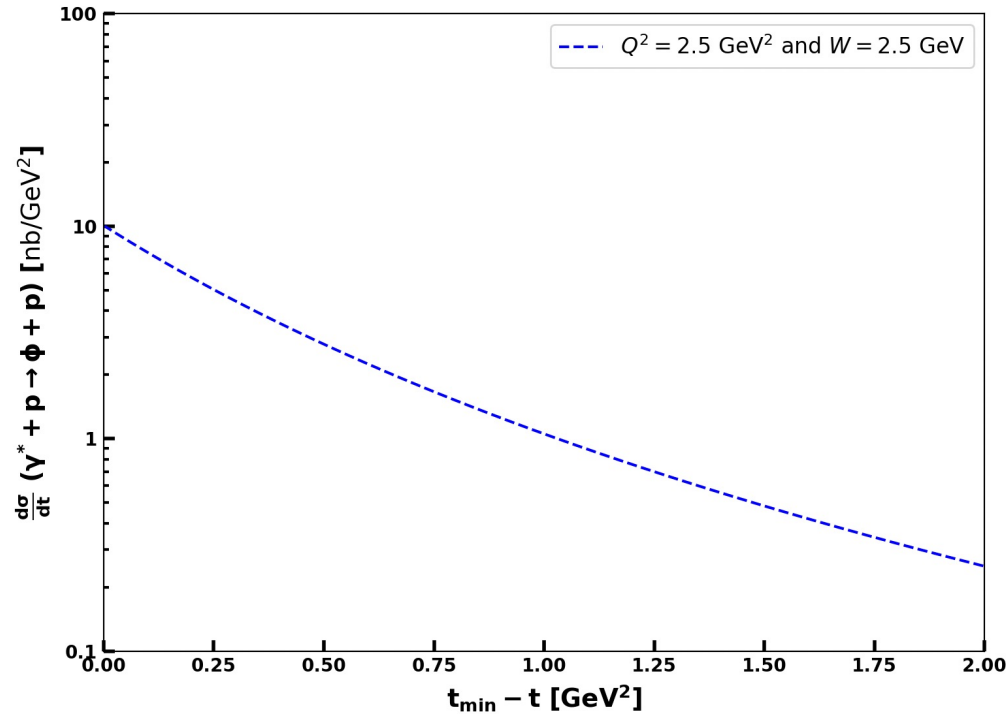
Cuts :

→ Cut on the **angle** between the **missing nucleon** and the **neutron**

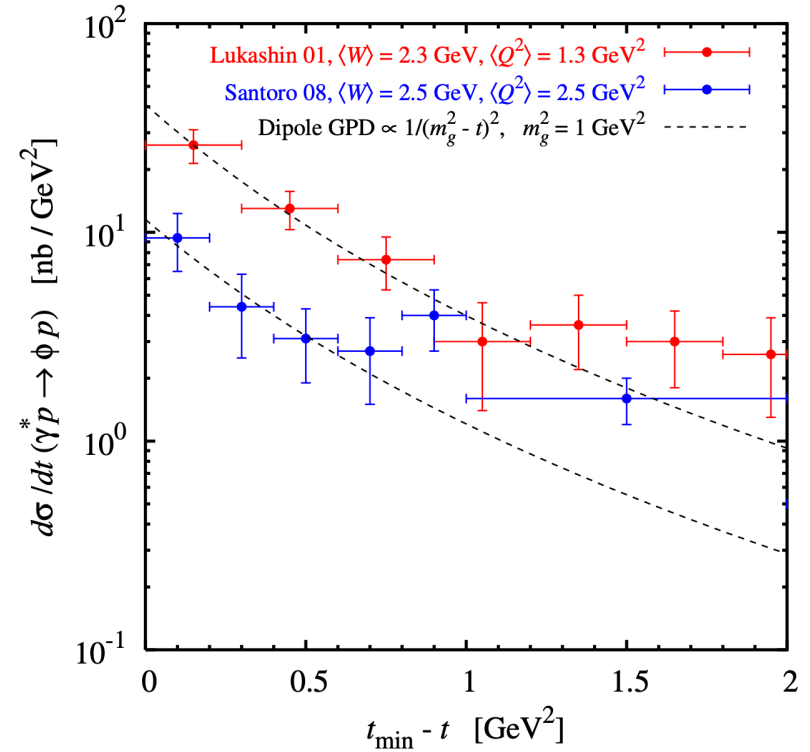
→ Cut on $-0.5 < \text{MissingMass}_{\text{tot}}^2 < 0.5 \text{ GeV}^2$

→ Cut on status of K^+ and K^- (keep event with both kaons in FD)

ϕ generator for MC simulation



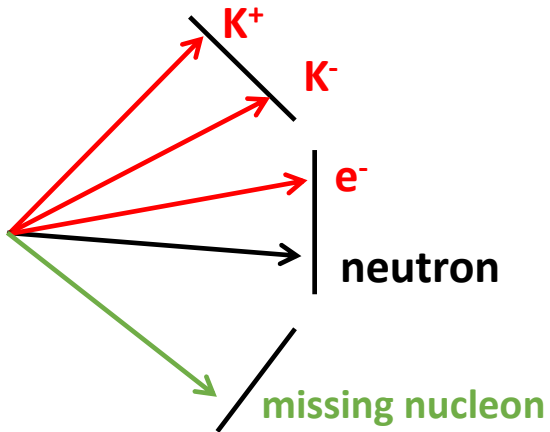
Implemented in the generator



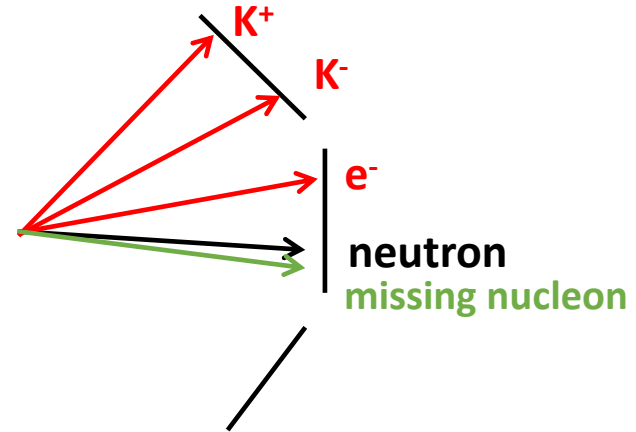
From Proposal to Jefferson Lab PAC39

Cut on the angle between missing nucleon and detected neutron

Missing nucleon = the missing particle of : $e n \rightarrow e' K^+ K^- X$



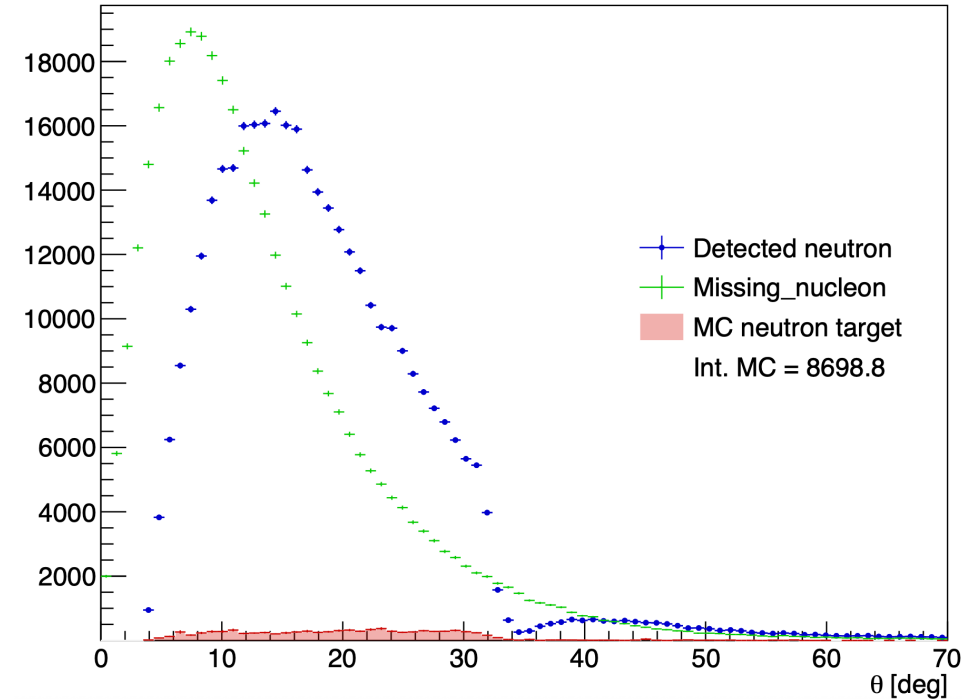
Not a good event



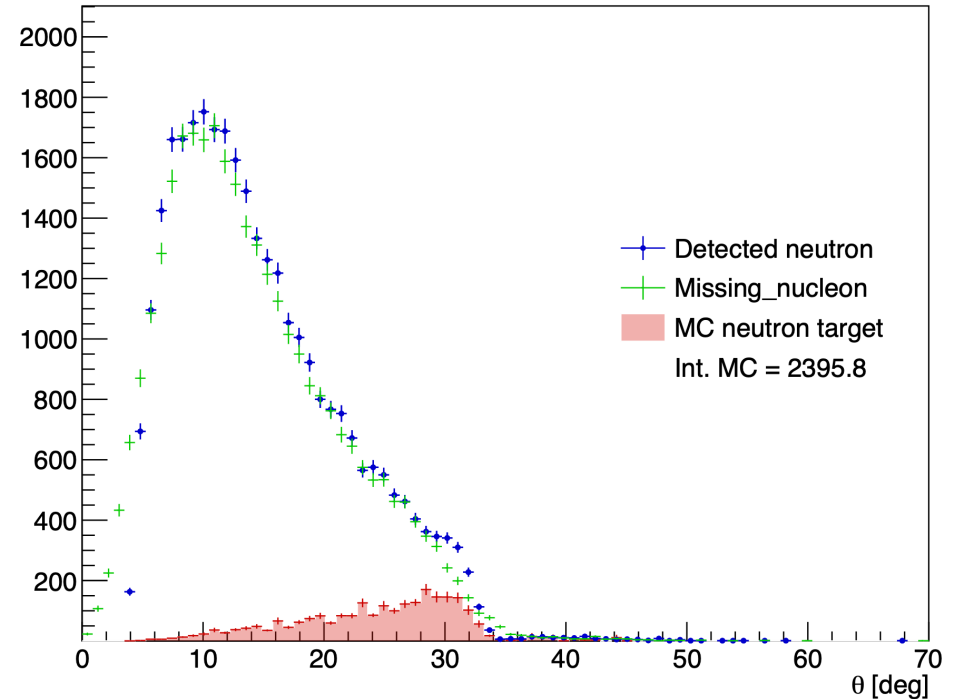
Good event

→ Cut on the angle between the missing nucleon and the neutron $< 5^\circ$

Cut on the angle between missing nucleon and detected neutron

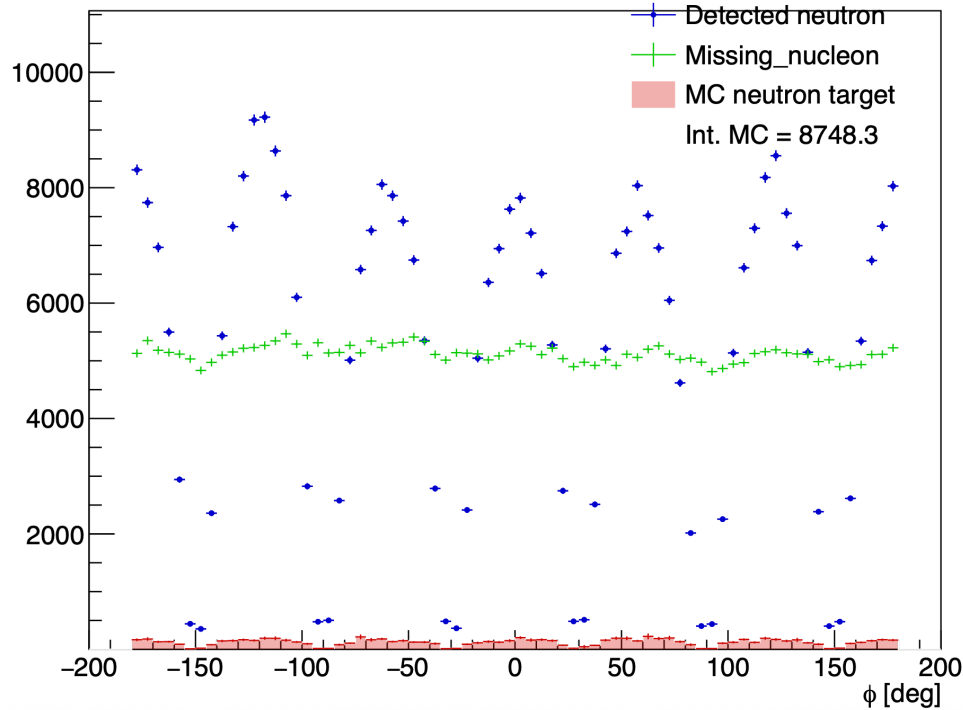


Without the angle cut

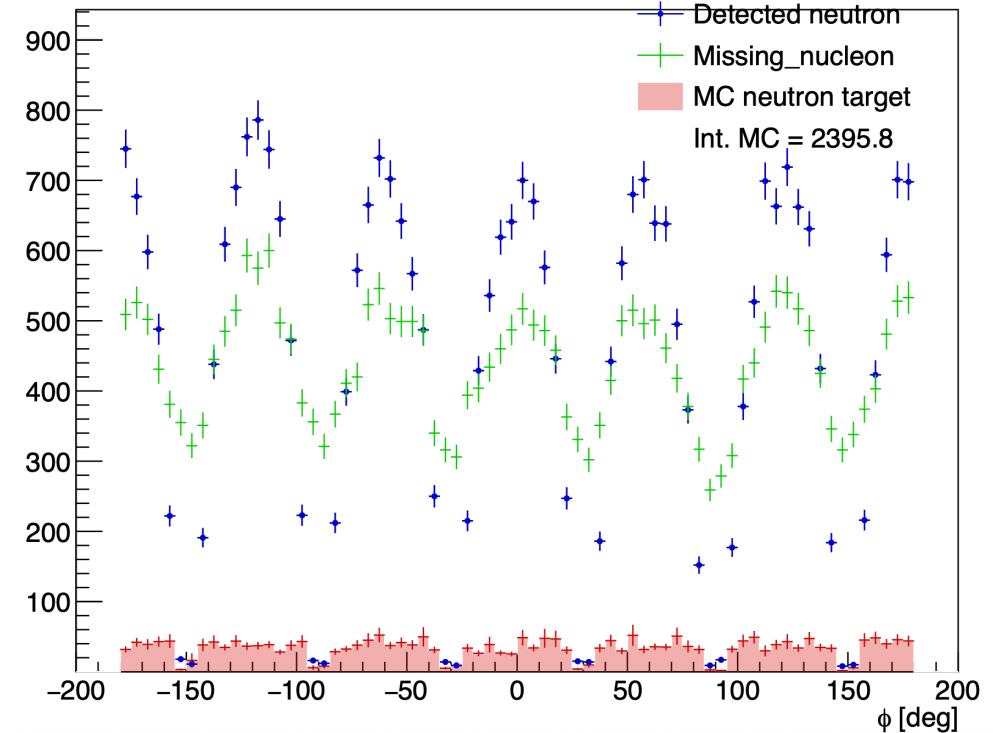


With the angle cut

Cut on the angle between missing nucleon and detected neutron

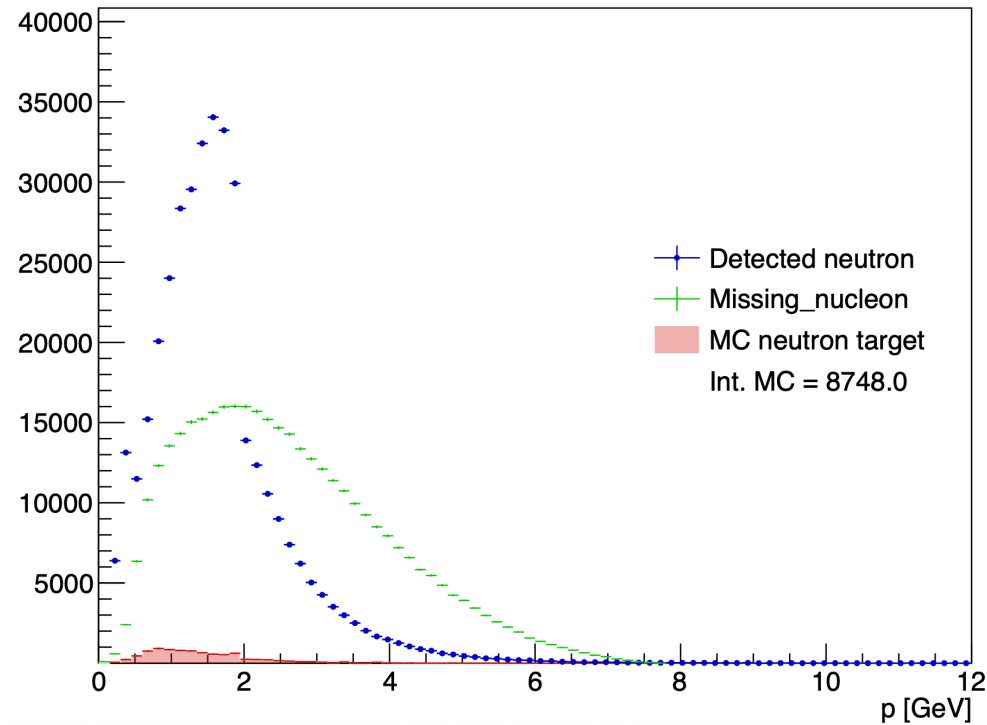


Without the angle cut

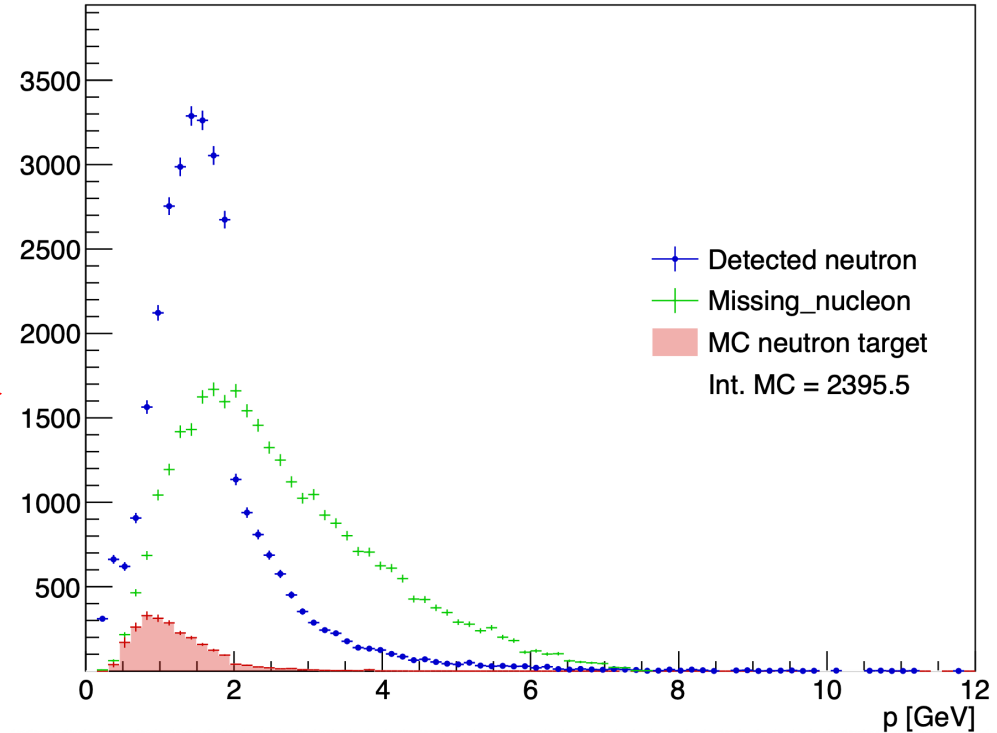


With the angle cut

Cut on the angle between missing nucleon and detected neutron

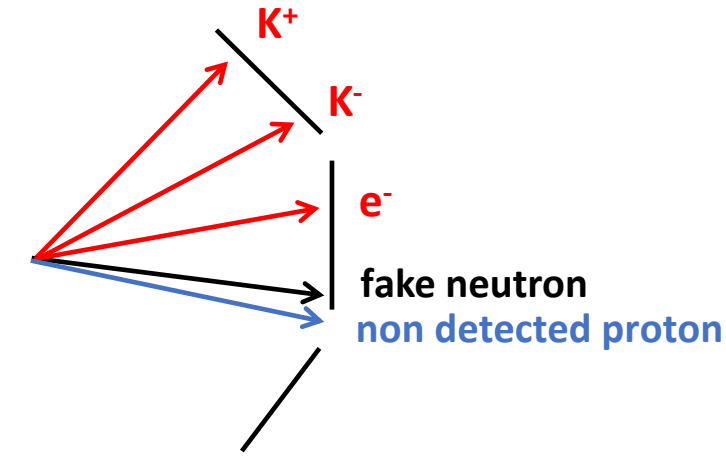


Without the angle cut

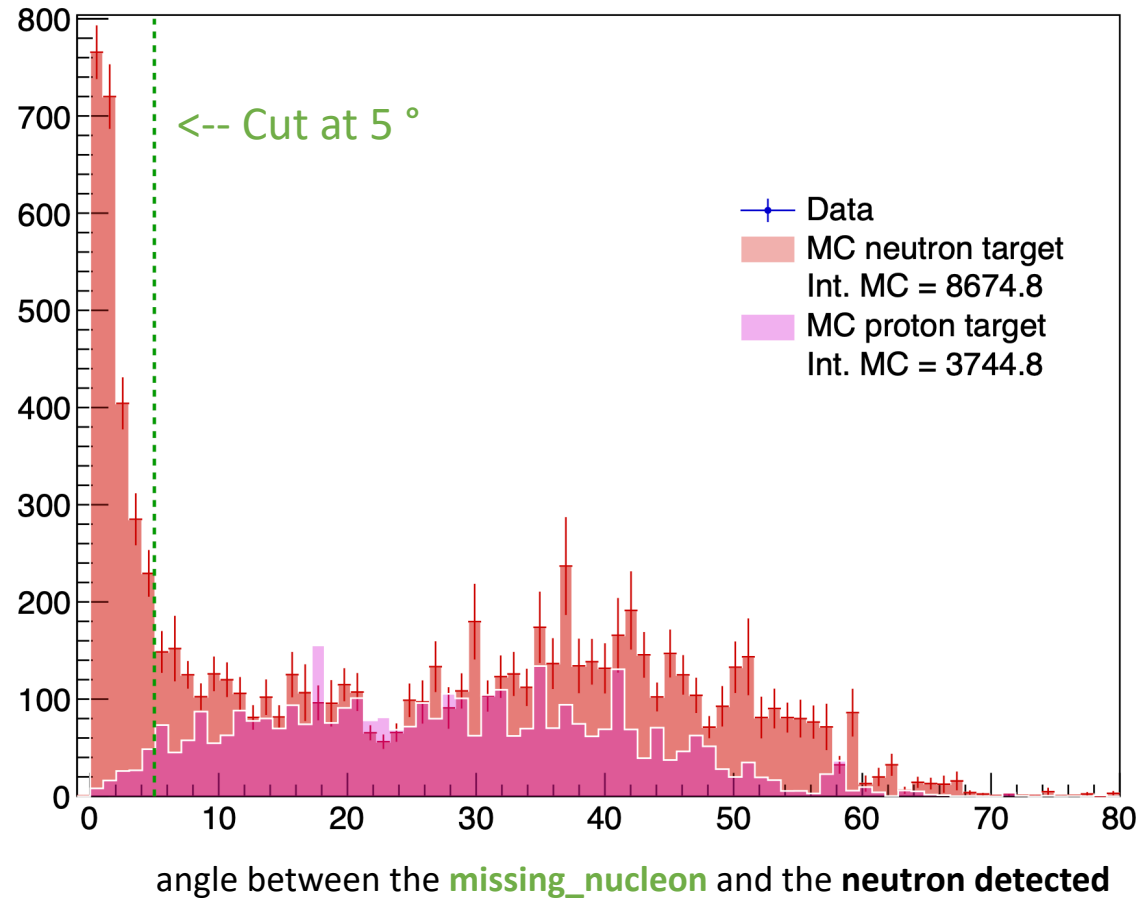


With the angle cut

Cut on the angle between missing nucleon and detected neutron



Contamination when the **missing_nucleon** is a **non detected proton** there is a fake neutron signal who pass the cut on the angle

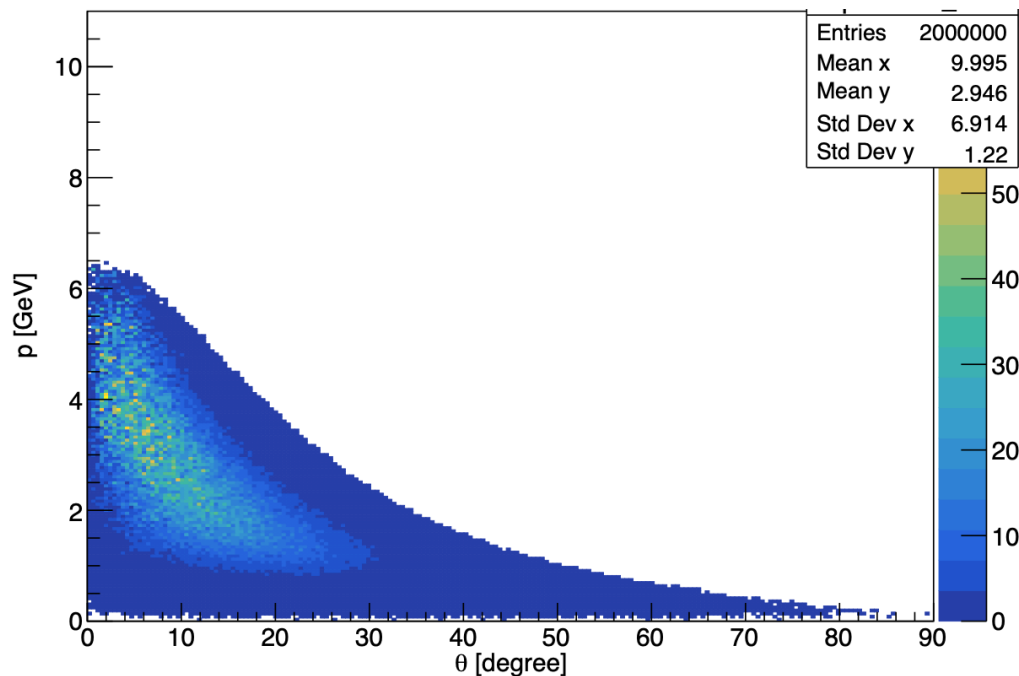


Cut on the status of K^+K^-

Two argument to cut $K^+ K^-$ in CD :

- Reconstruction of kaons in the CD is less good than in the FD
- The generator predict few events in the CD

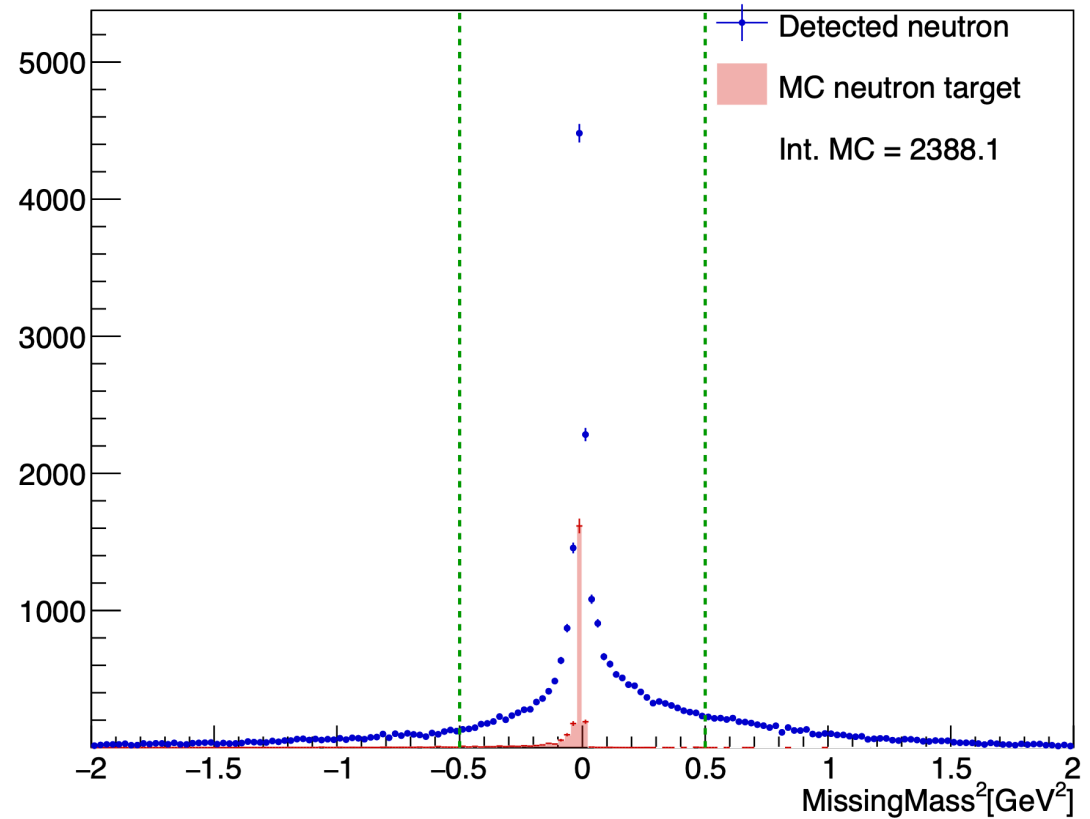
Generated event : p vs theta for K^+



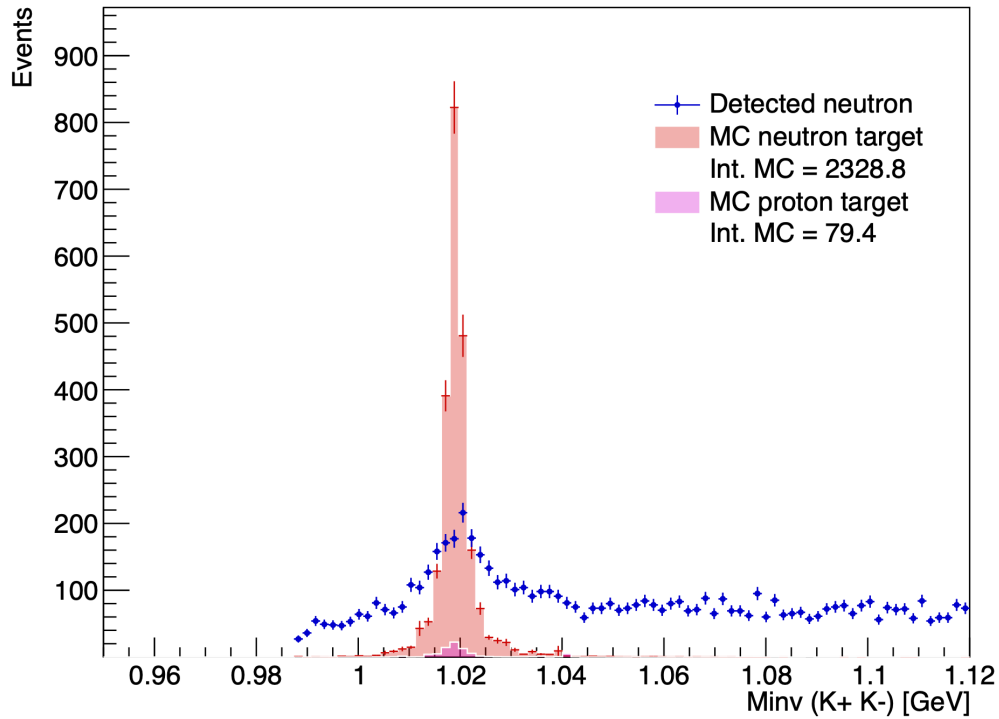
Cut on total missing mass

Total missing mass $e n \rightarrow e' n' K^+ K^- X$
(after all the previous cut)

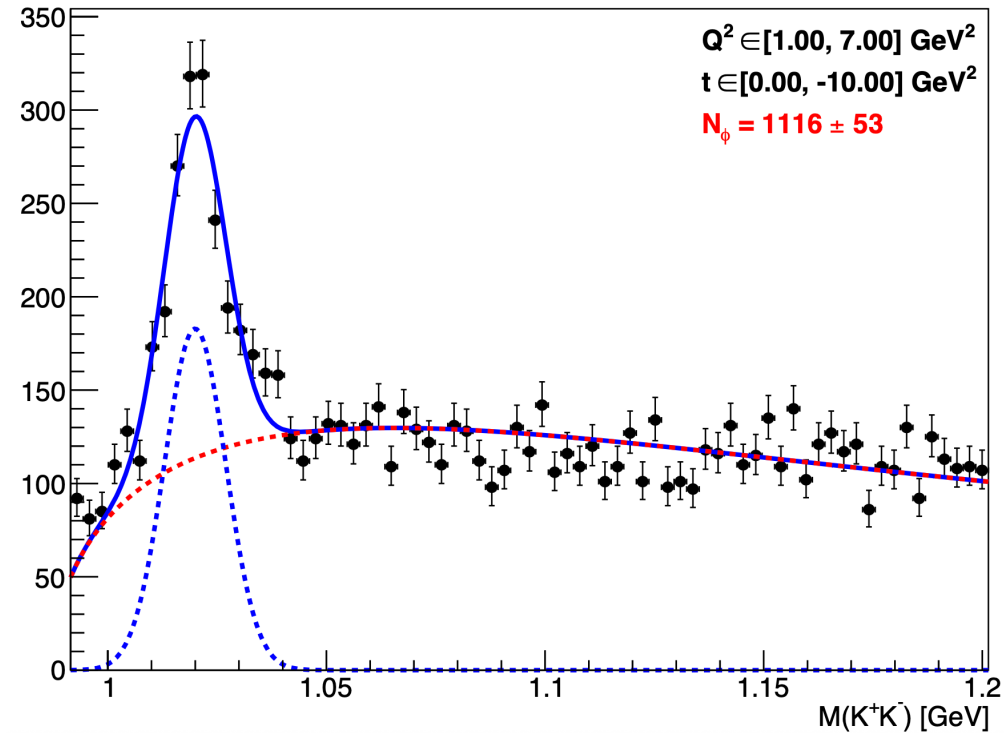
$$-0.5 < \text{MissingMass}_{\text{tot}}^2 < 0.5 \text{ GeV}^2$$



Yield extraction

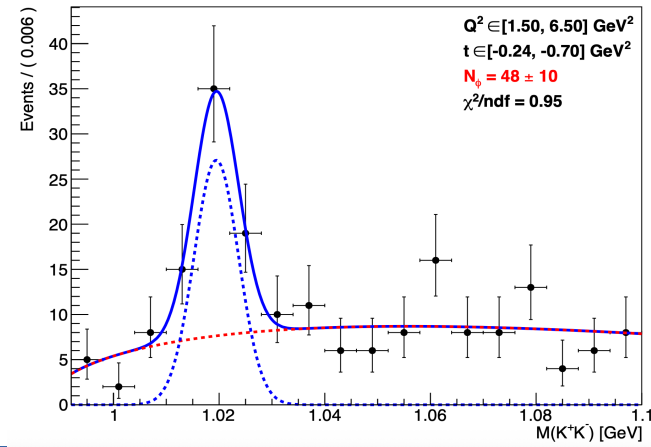
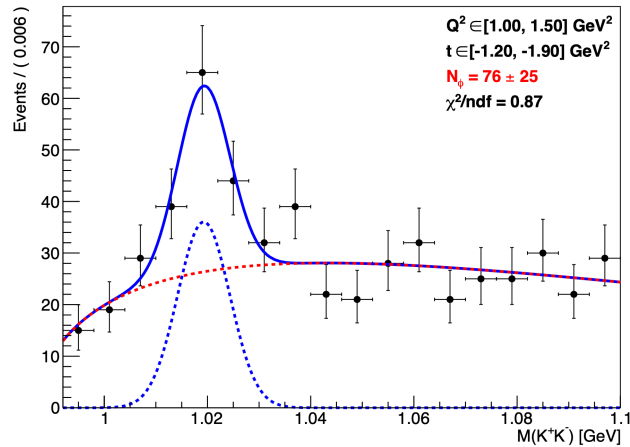
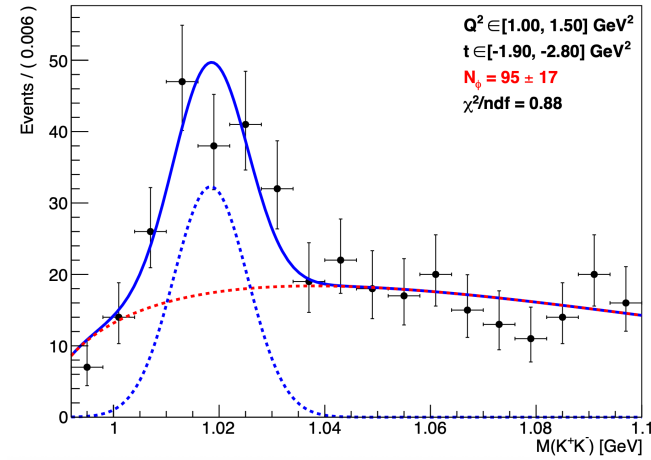
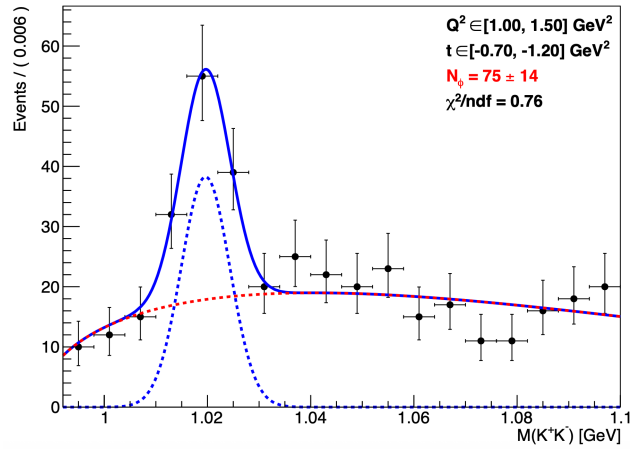


With angle cut, $K^+ K^-$ in FD, and cut on MissingMass



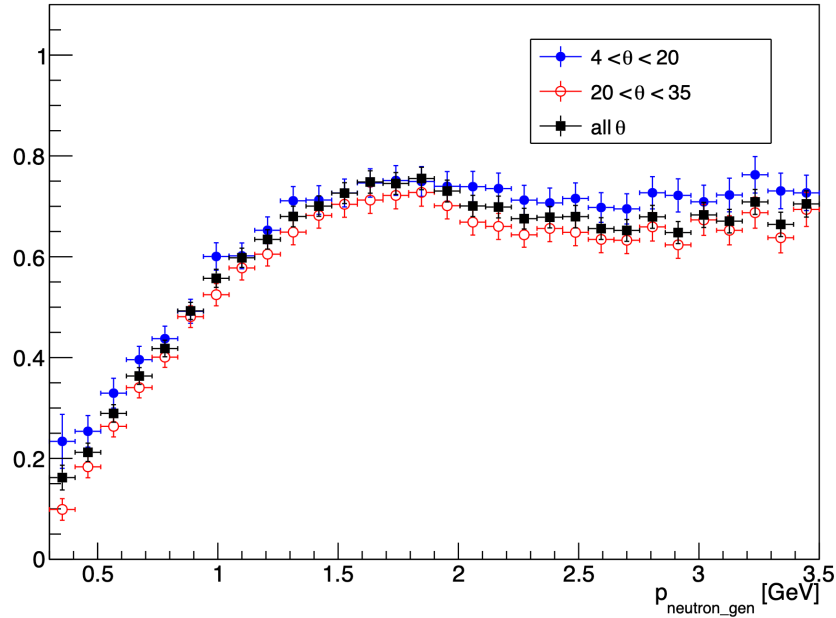
Fit all fall2019 outbending data

Yield extraction



Efficiency correction

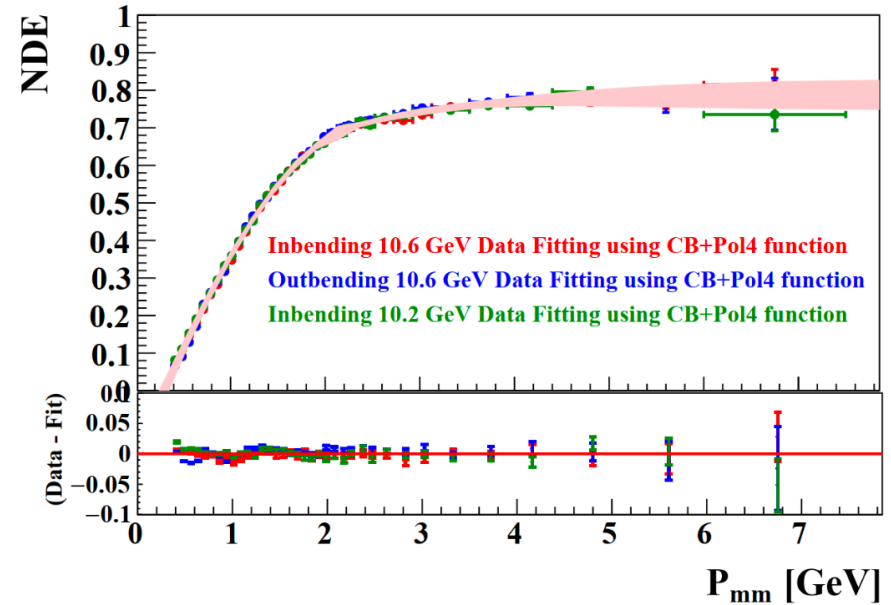
MC Efficiency vs p



Neutron efficiency in MC :

(Fiducial cut for Nrec and acceptance cut for Ngen)

Neutron Detection Eff

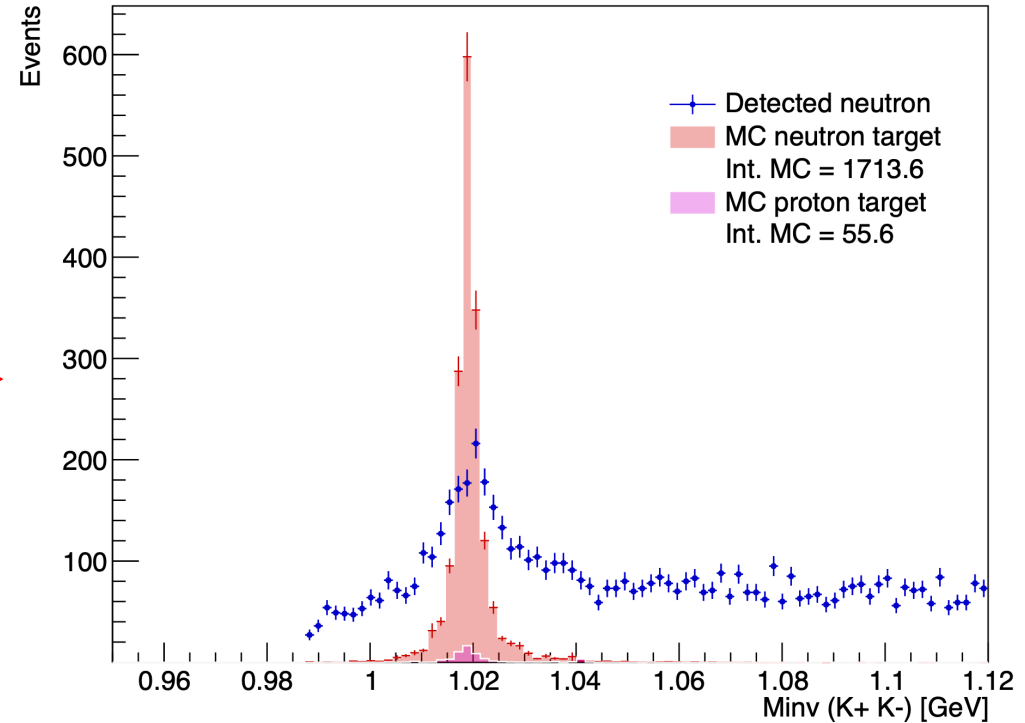
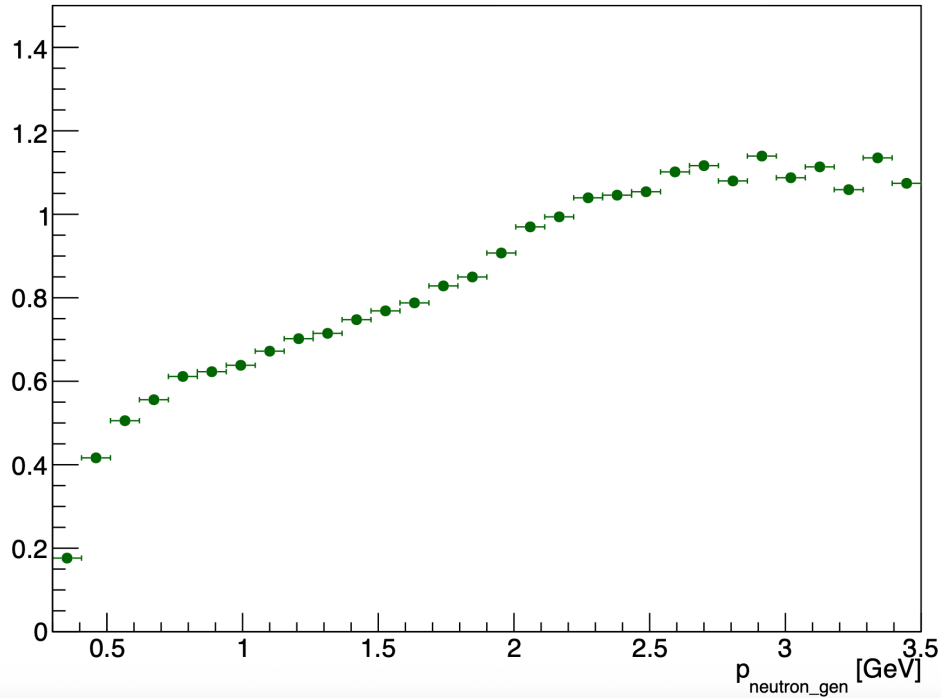


Neutron efficiency (FD) in data from **Lamya Baashen** thesis

(with P_{mm} the missing momentum)

Efficiency correction

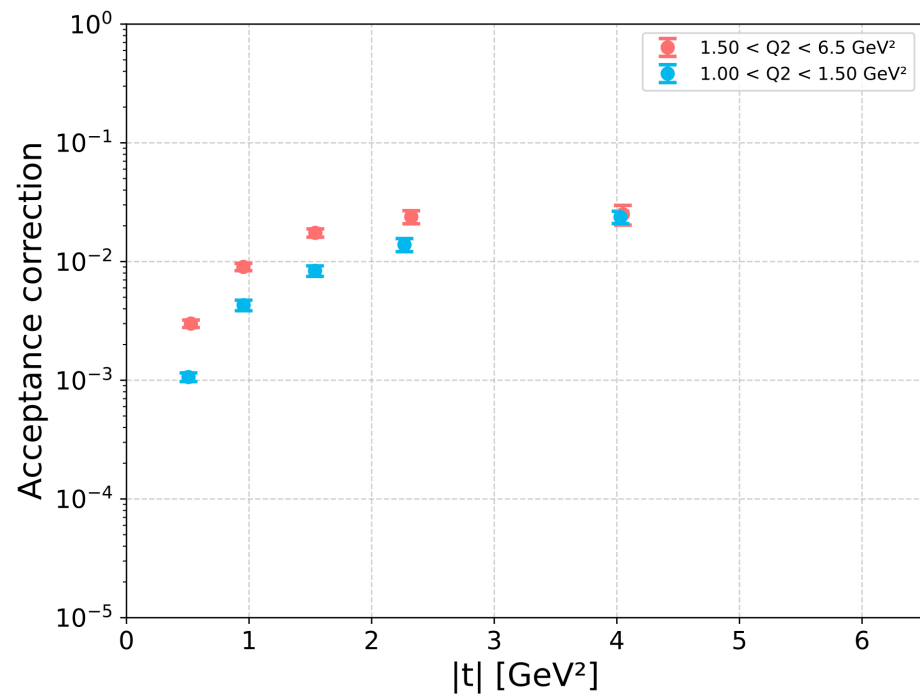
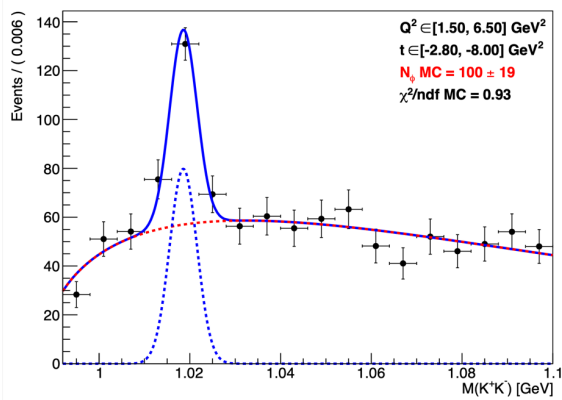
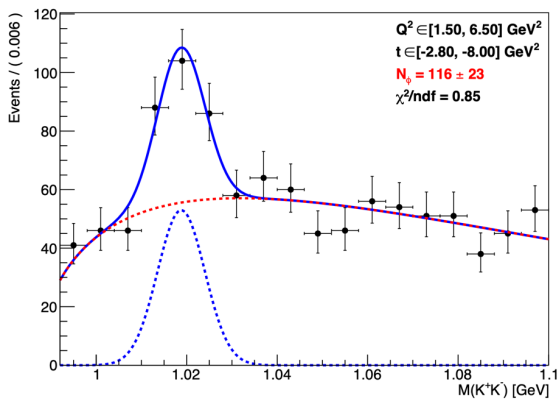
Data/MC Efficiency Ratio



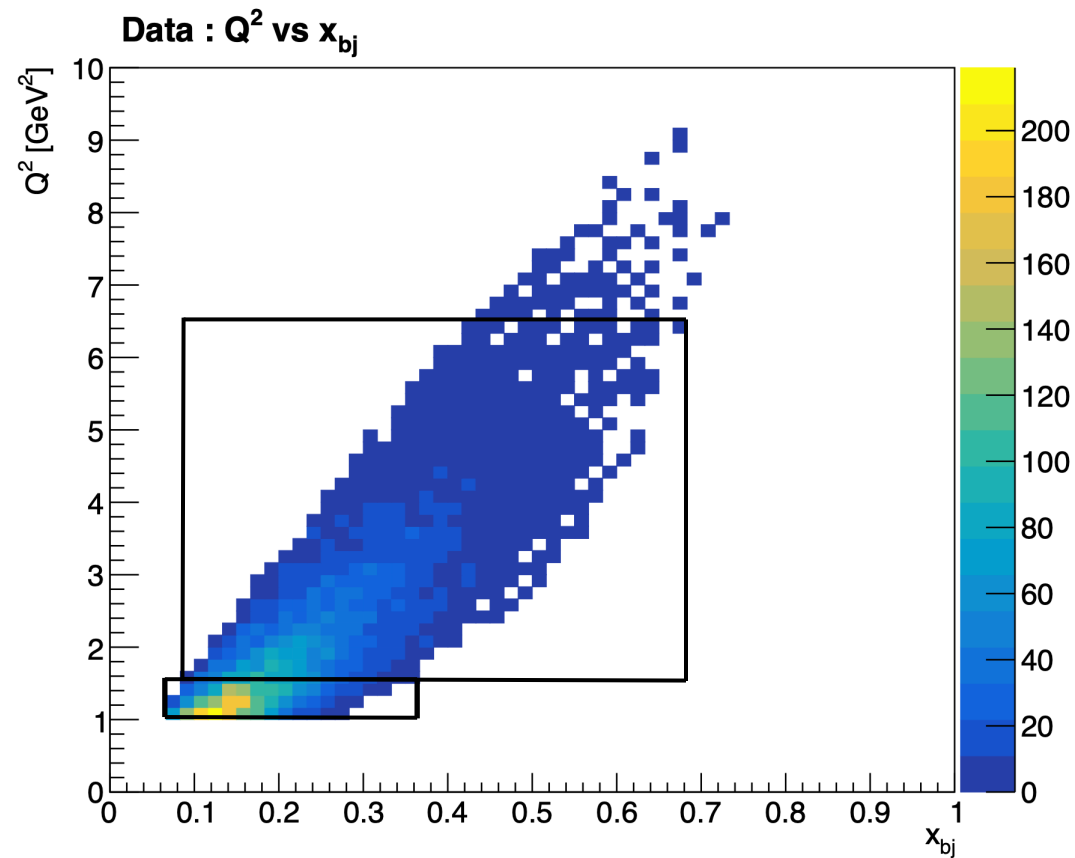
MC with neutron efficiency correction

Acceptance correction

A =



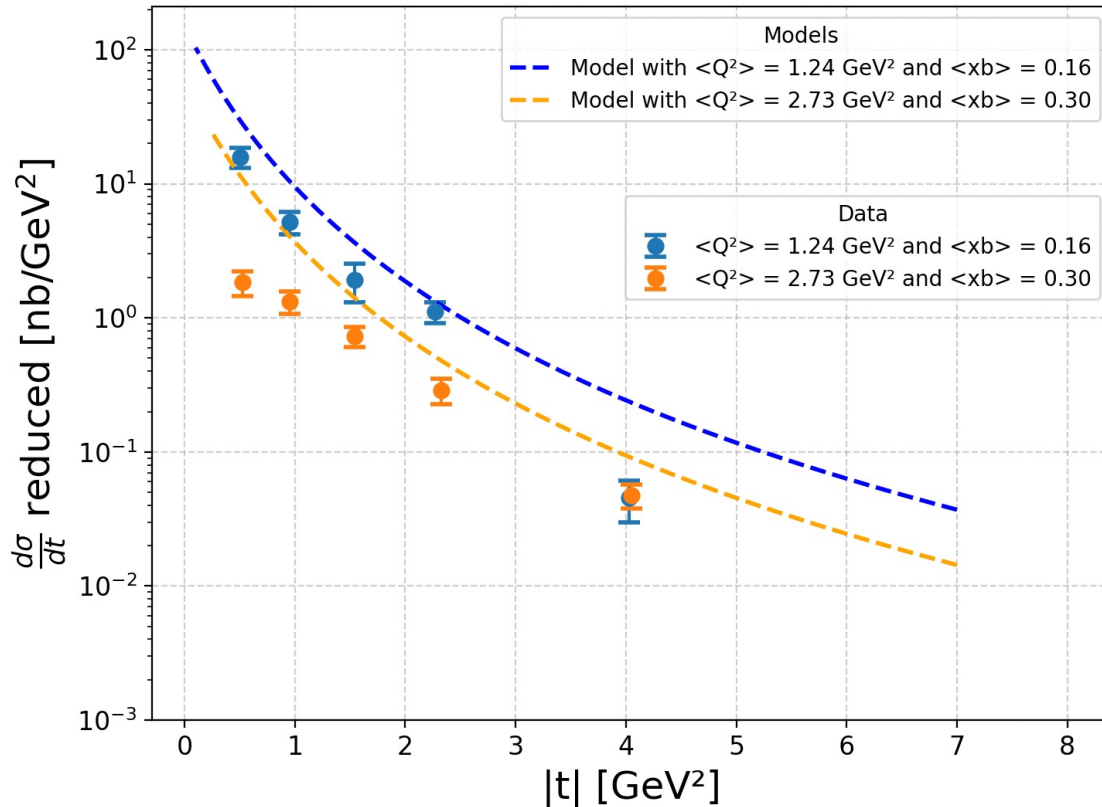
Binning correction



$f =$

bin	$1.0 < Q^2 < 1.5 \text{ GeV}^2$ $0.05 < x_b < 1.34$	$1.5 < Q^2 < 6.5 \text{ GeV}^2$ $0.08 < x_b < 0.7$
f	0.81	0.49

Preliminary plot of the differential reduced cross section



$$\frac{d^3\sigma}{dQ^2 dx_B dt} = \Gamma(Q^2, x_B, E) \left[\frac{d\sigma_T}{dt}(Q^2, x_B, t) + \epsilon \frac{d\sigma_L}{dt}(Q^2, x_B, t) \right]$$

$$\Gamma \equiv \frac{\alpha}{8\pi} \frac{Q^2}{m_N^2 E^2} \frac{1-x_B}{x_B^3} \frac{1}{1-\epsilon}.$$

$$\frac{d^3\sigma}{dQ^2 dx_B dt} = \frac{\Delta N}{\Delta Q^2 \cdot \Delta x_B \cdot \Delta t \cdot L \cdot \text{Br} \cdot \text{Acc.}}$$

Conclusion

Next steps :

- Improve fits for signal and background
- Run the code with the other data set
- More MC simulations
- Events in CD
- Momentum corrections

Thanks!

Backup slides

ϕ generator for MC simulation

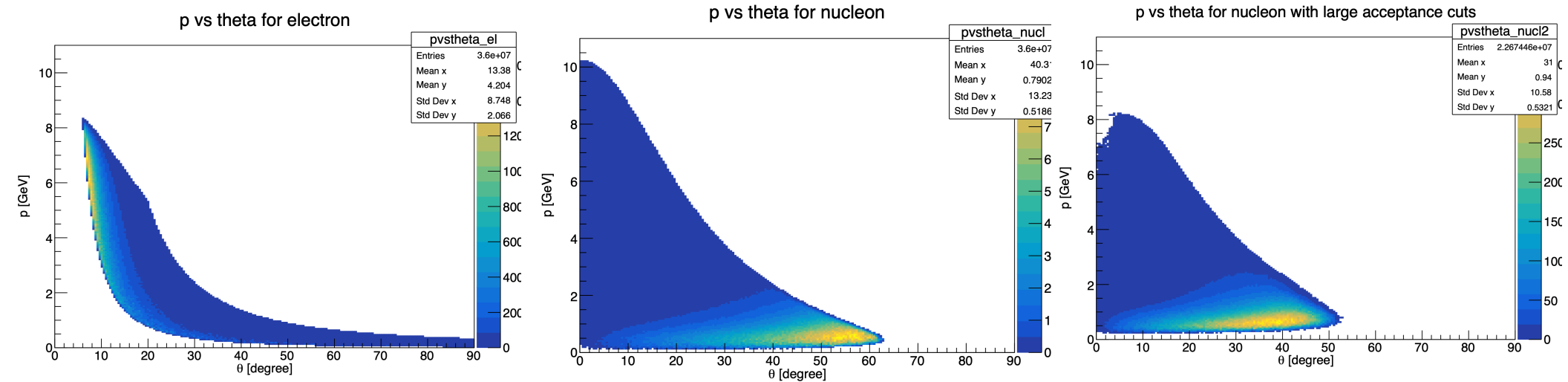
$$weight_{PhaseSpace} = |Q_{max}^2 - Q_{min}^2| * |xb_{max} - xb_{min}| * |t_{max} - t_{min}|$$

$$\frac{d^3\sigma}{dQ^2 dx_B dt} \quad \text{From Proposal to Jefferson Lab PAC39 Exclusive Phi Meson Electroproduction with CLAS12}$$

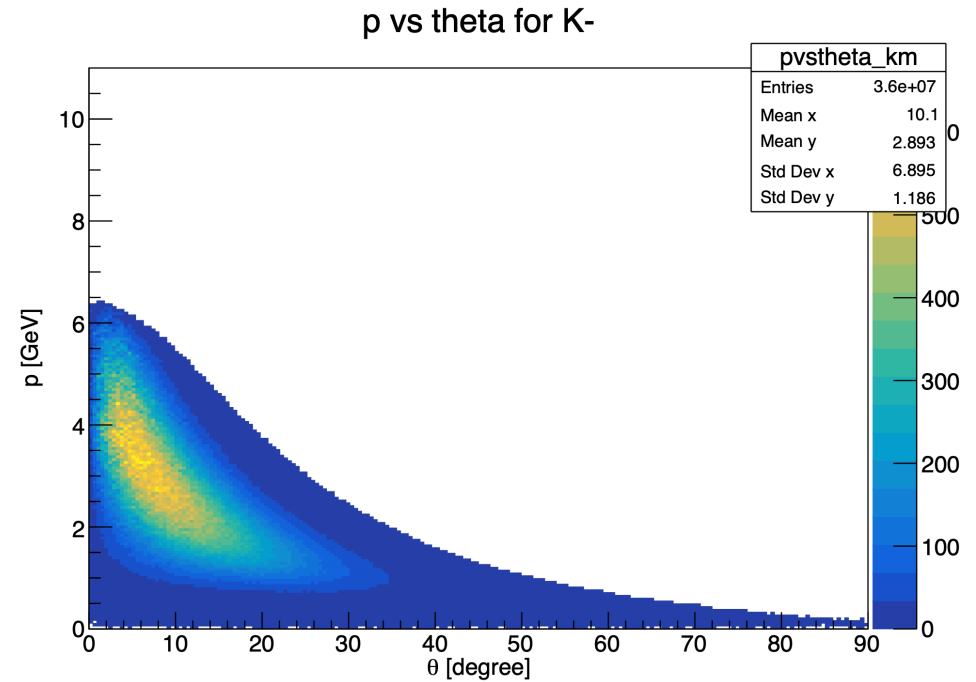
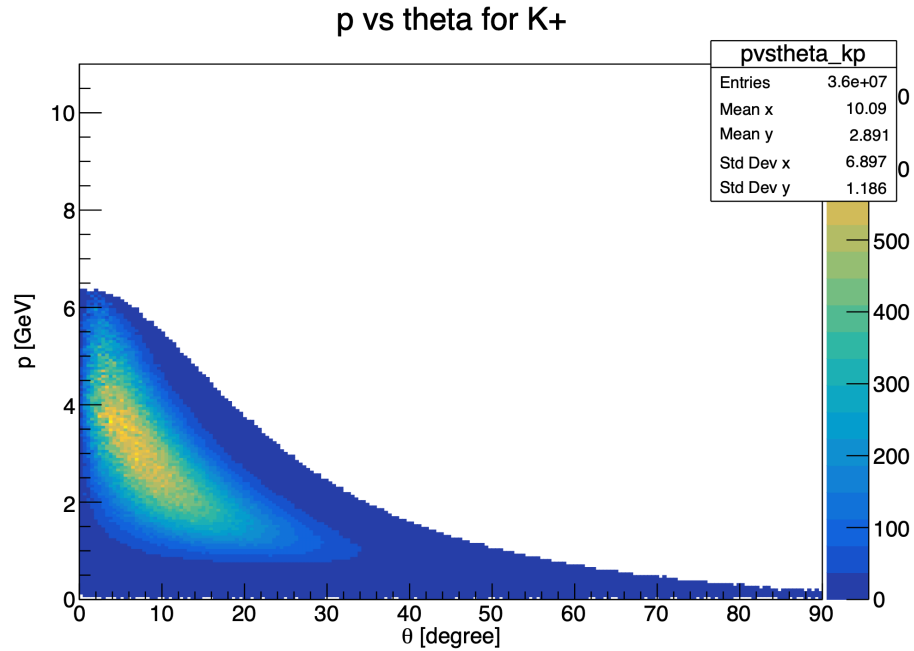
$$BR(\phi \rightarrow K^+ K^-) \approx 49.2\%$$

$$totalweight = weight_{phasespace} * weight_{\frac{d^3\sigma}{dt dQ^2 dx_b}} * BR_{\phi \rightarrow K^+ K^-}$$

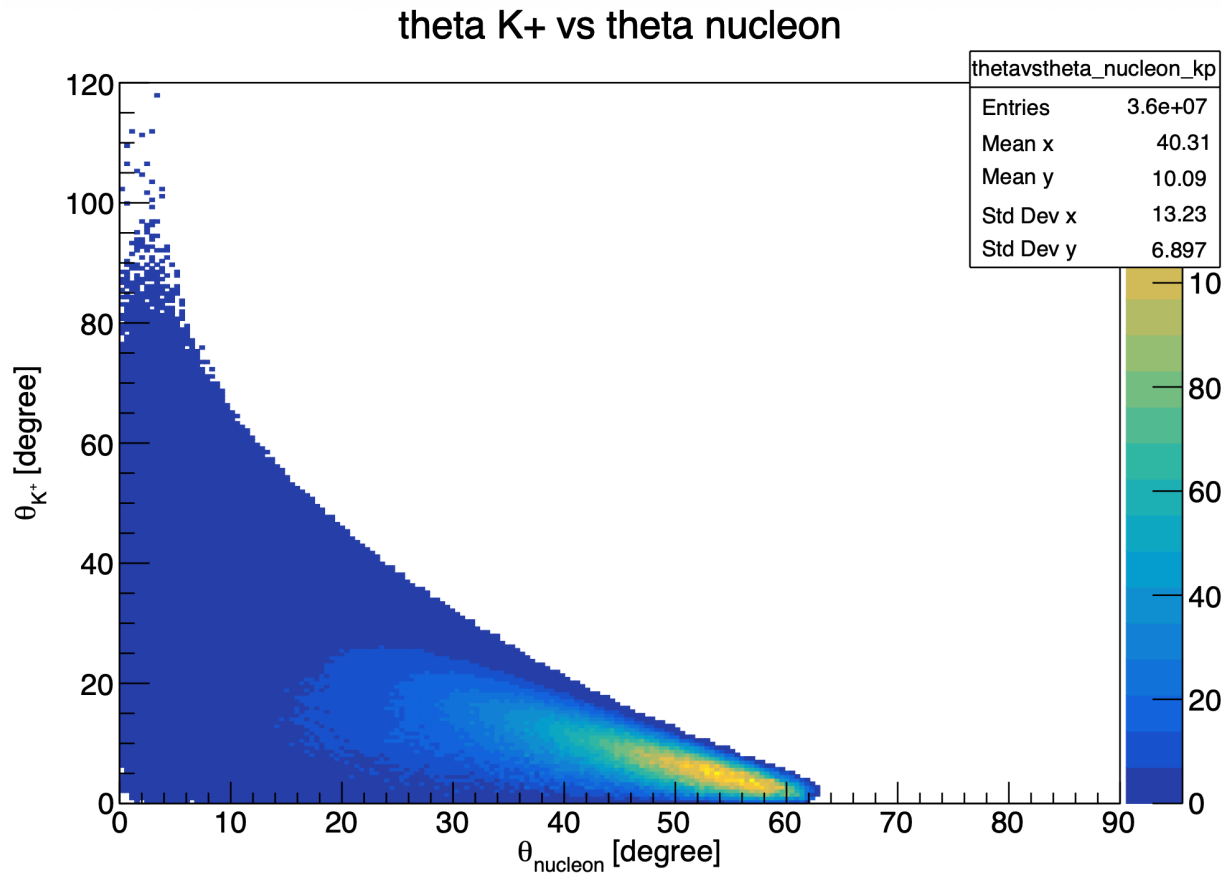
Generator plots



Generator plots

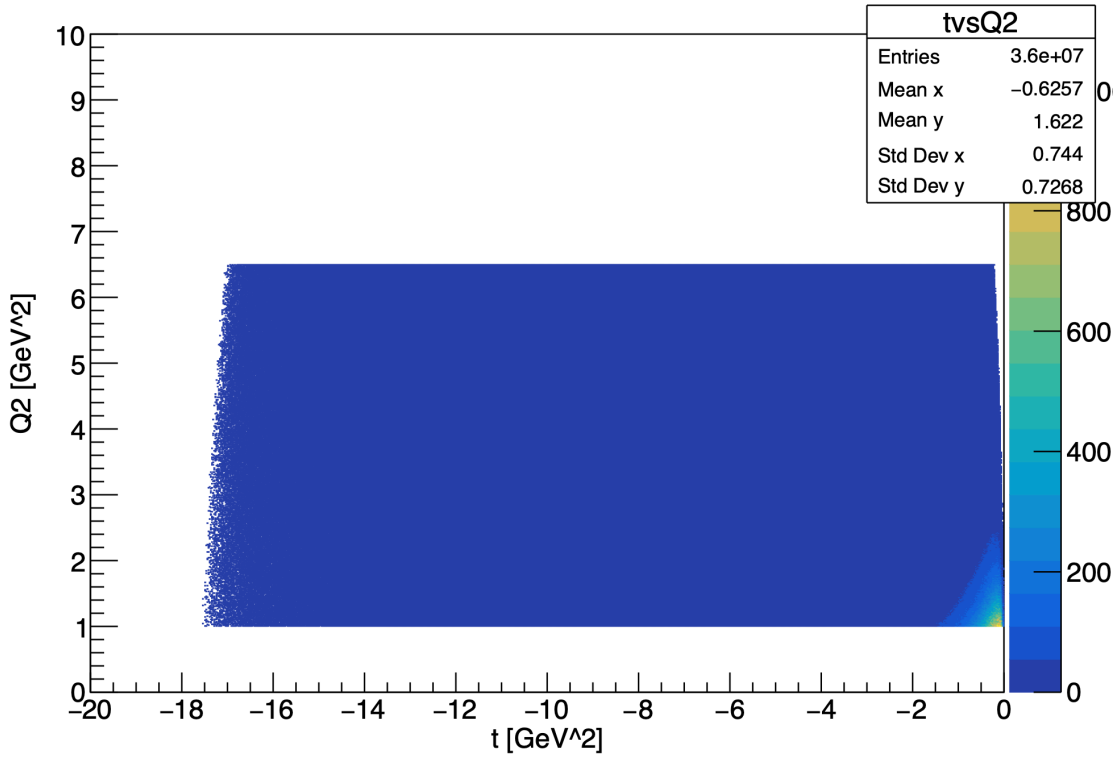


Generator plots

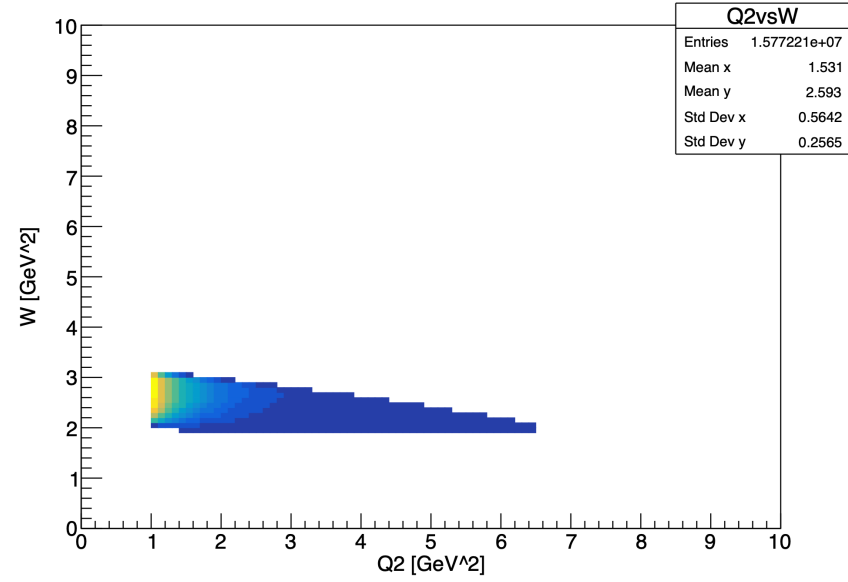


Generator plots

t vs Q2 with kinematics cuts

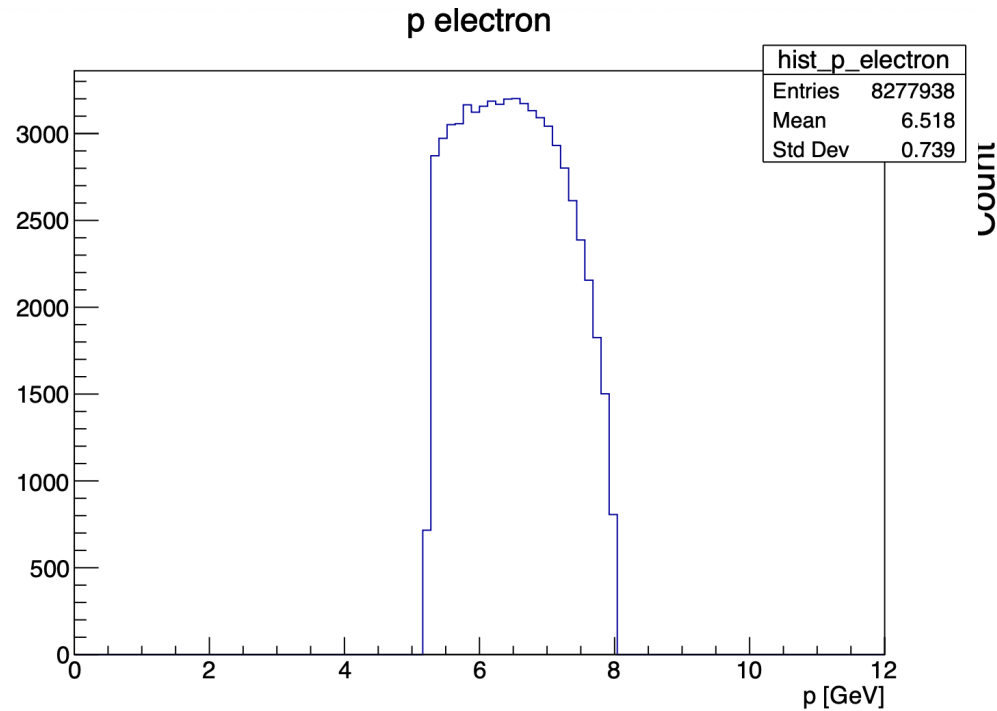


Q2 vs W

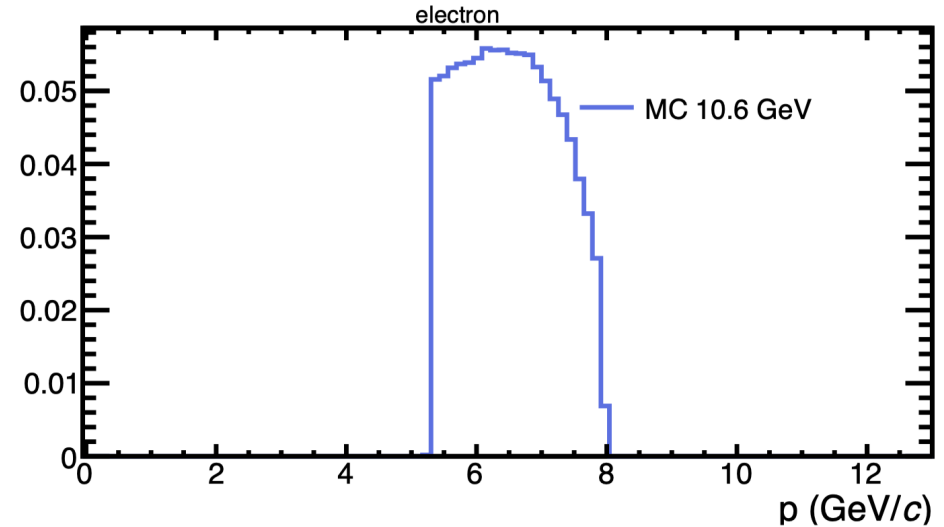


Generator plots

With this cuts : $5.25 < P_{\text{electron}} < 8 \text{ GeV}$ and $P_{\text{proton}} > 0.45 \text{ GeV}$



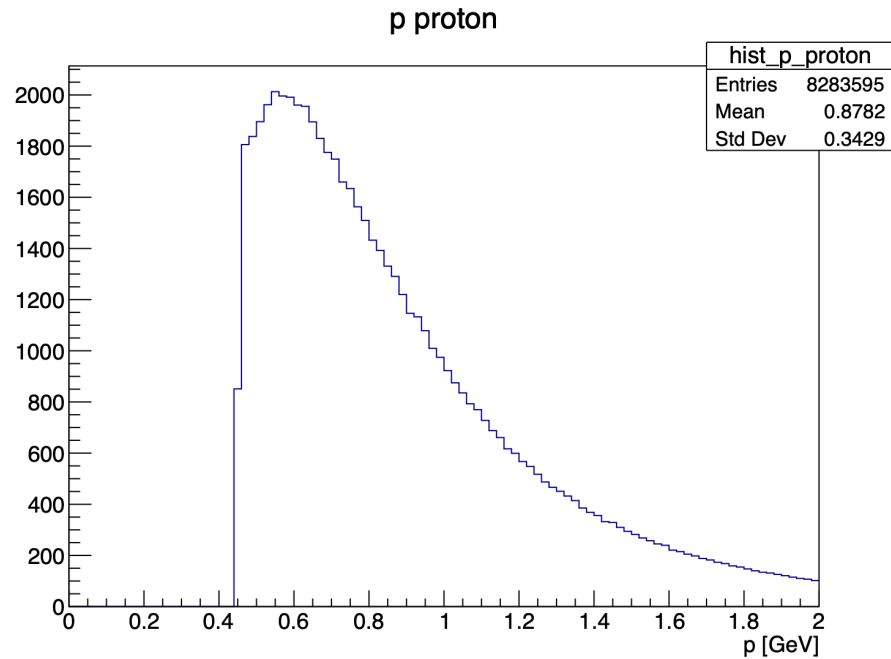
my generator



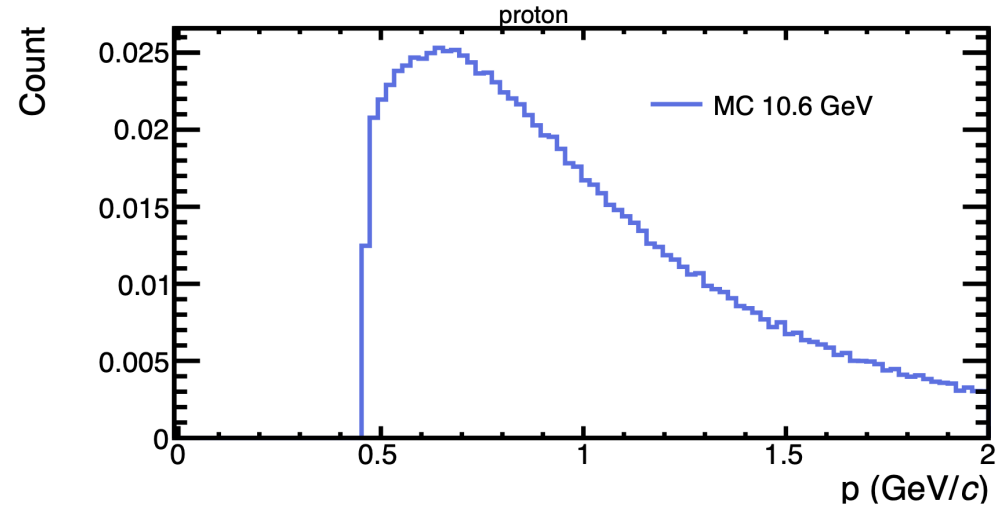
Bhawani plot

Generator plots

With this cuts : $5.25 < P_{\text{electron}} < 8 \text{ GeV}$ and $P_{\text{proton}} > 0.45 \text{ GeV}$



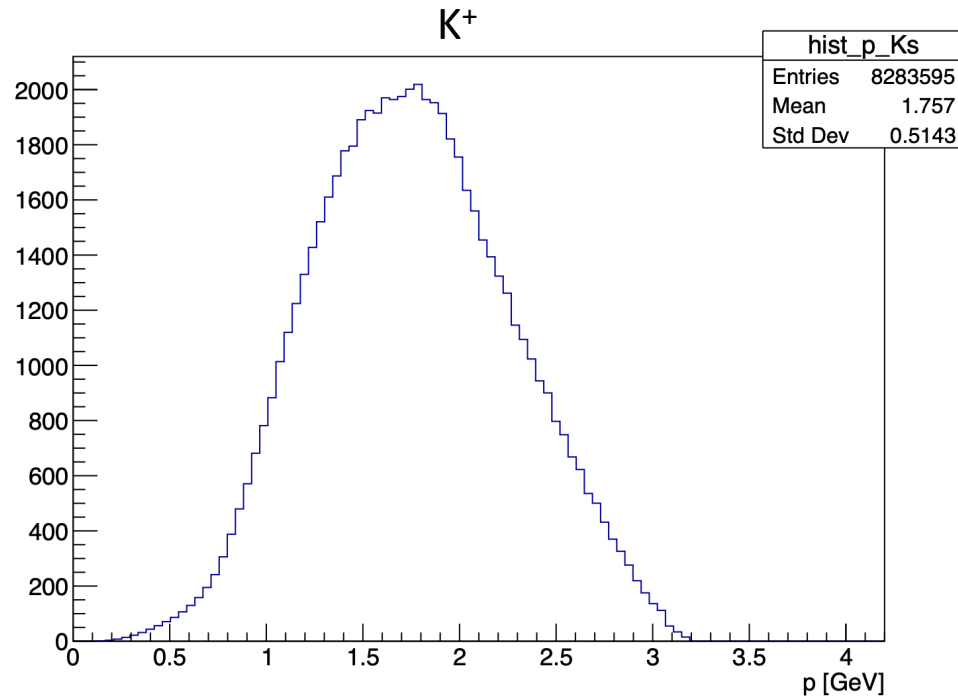
my generator



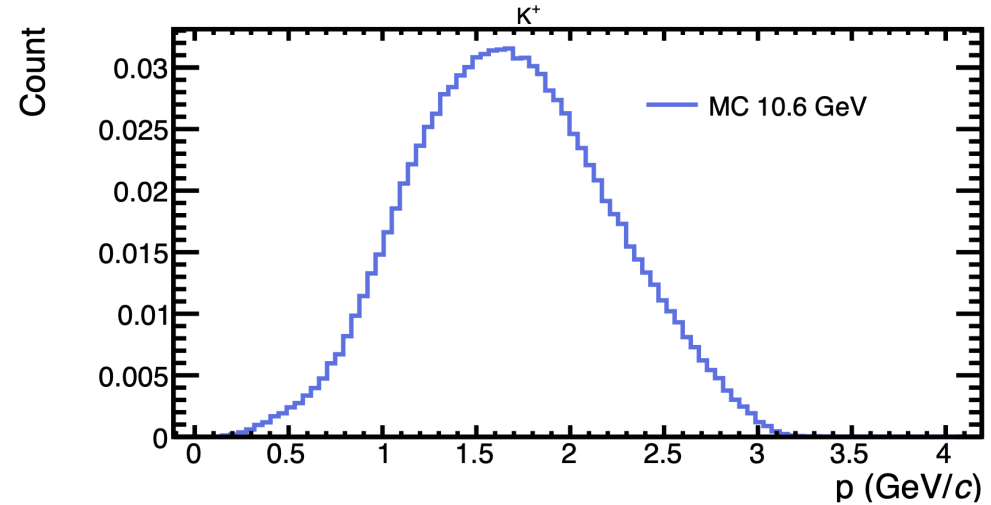
Bhawani plot

Generator plots

With this cuts : $5.25 < P_{\text{electron}} < 8 \text{ GeV}$ and $P_{\text{proton}} > 0.45 \text{ GeV}$



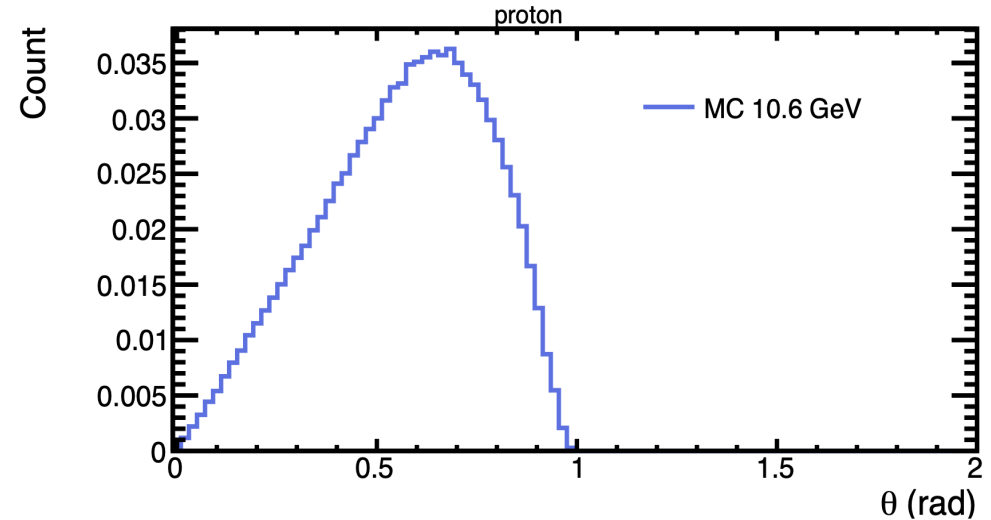
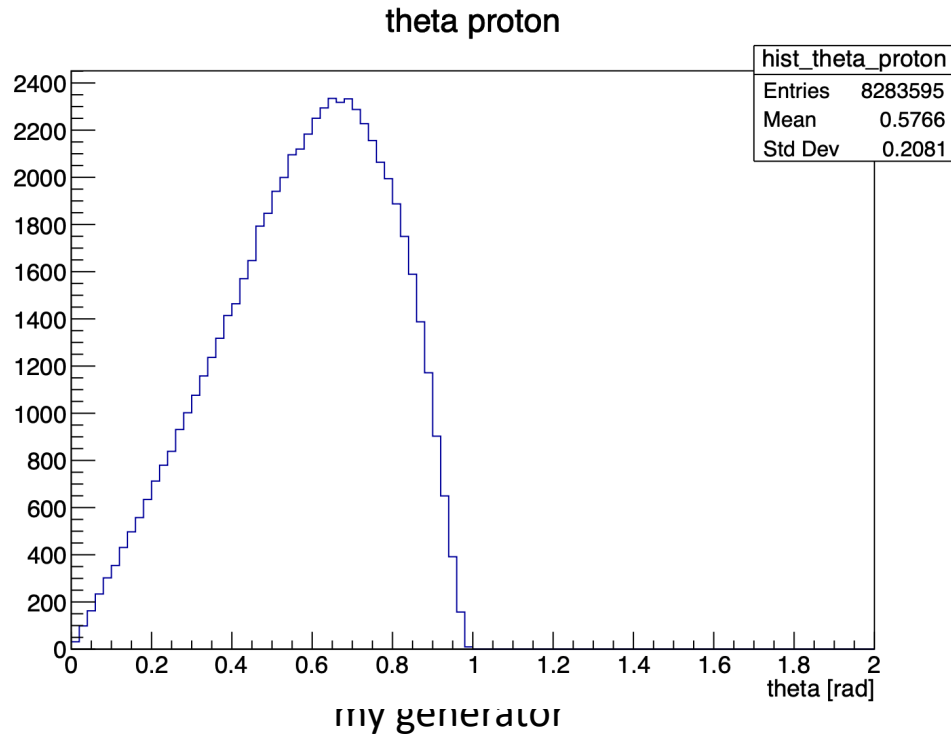
my generator



Bhawani plot

Generator plots

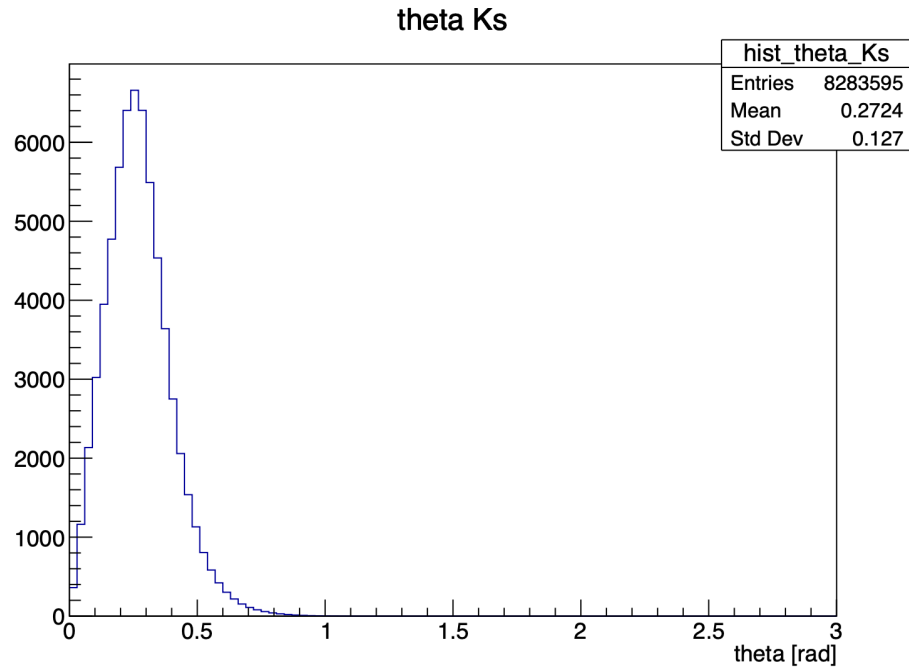
With this cuts : $5.25 < P_{\text{electron}} < 8 \text{ GeV}$ and $P_{\text{proton}} > 0.45 \text{ GeV}$



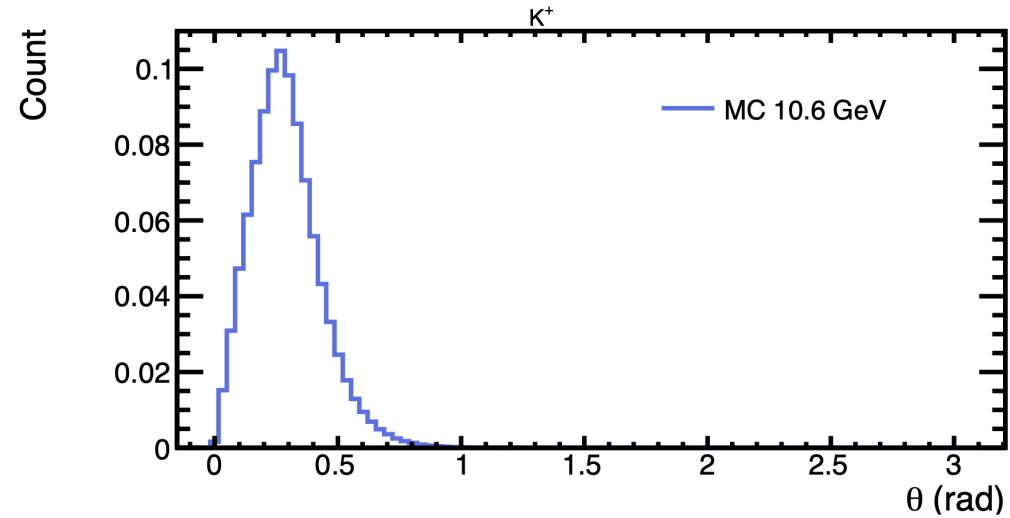
Bhawani plot

Generator plots

With this cuts : $5.25 < P_{\text{electron}} < 8 \text{ GeV}$ and $P_{\text{proton}} > 0.45 \text{ GeV}$



my generator

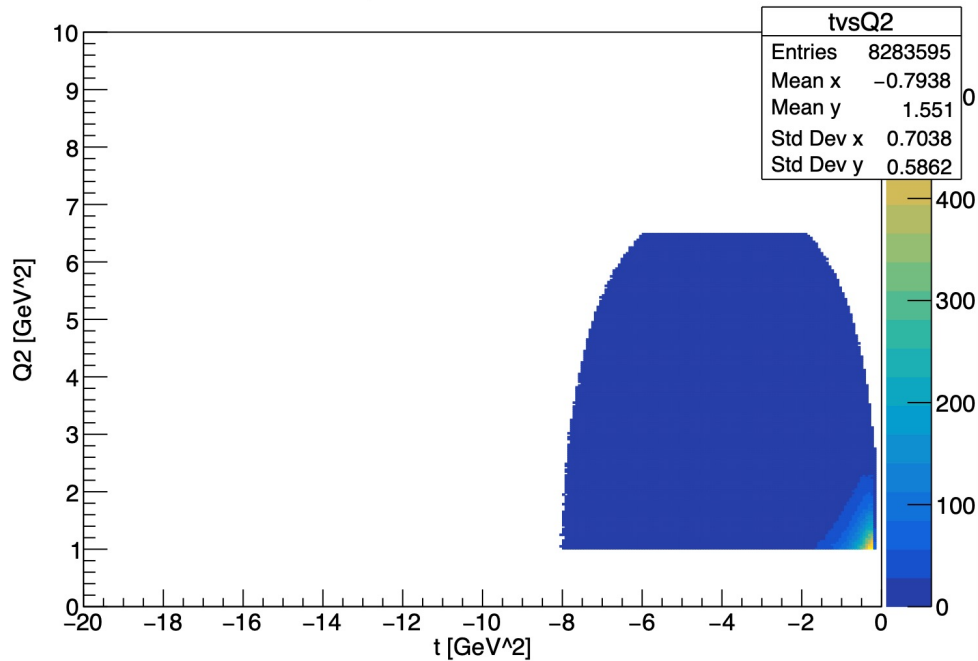


Bhawani plot

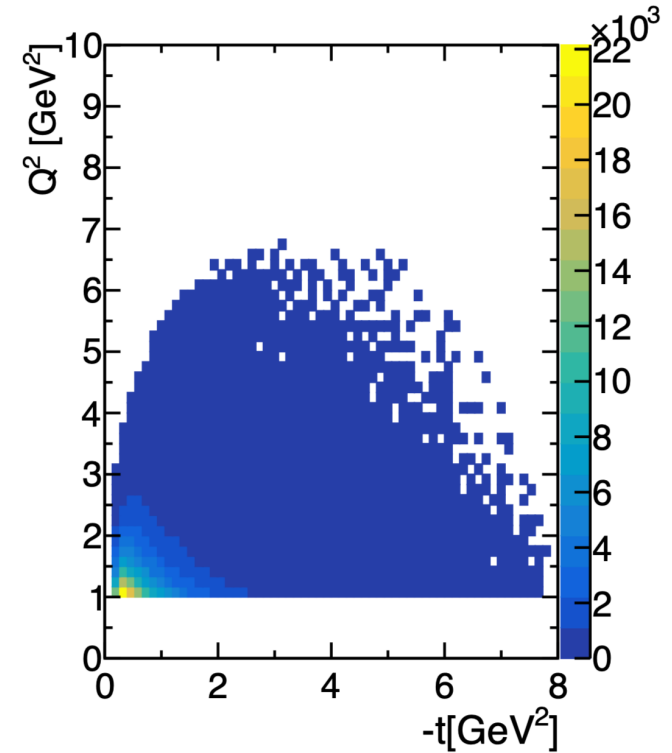
Generator plots

With this cuts : $5.25 < P_{\text{electron}} < 8 \text{ GeV}$ and $P_{\text{proton}} > 0.45 \text{ GeV}$

t vs Q2 with kinematics cuts



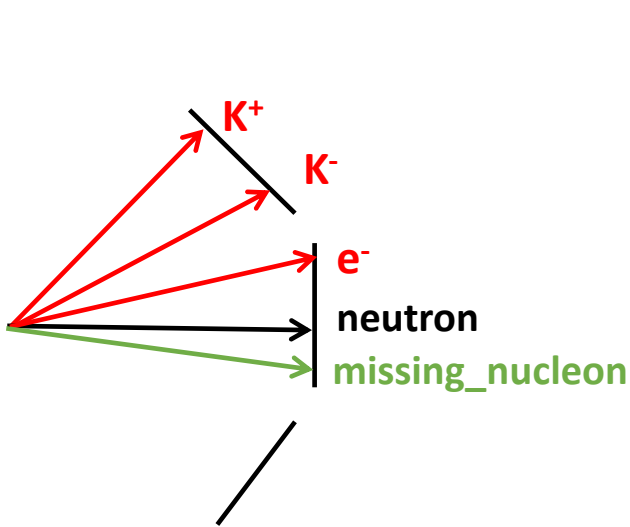
my generator



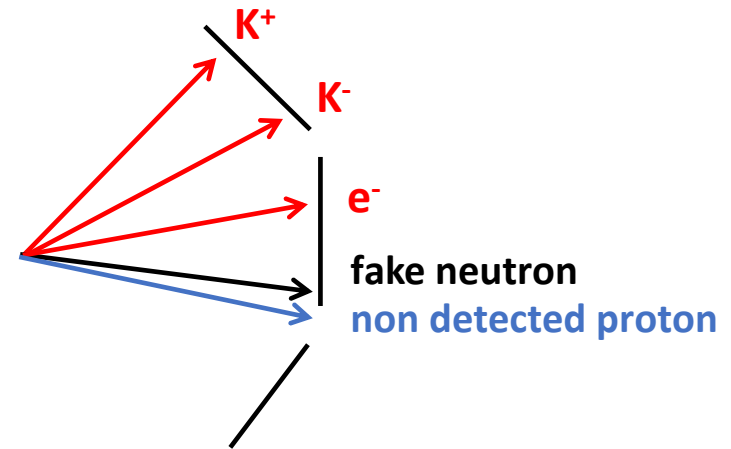
Bhawani plot

Proton contamination

Missing_nucleon = the missing particle of this reaction : $e n \rightarrow e' K^+ K^- X$



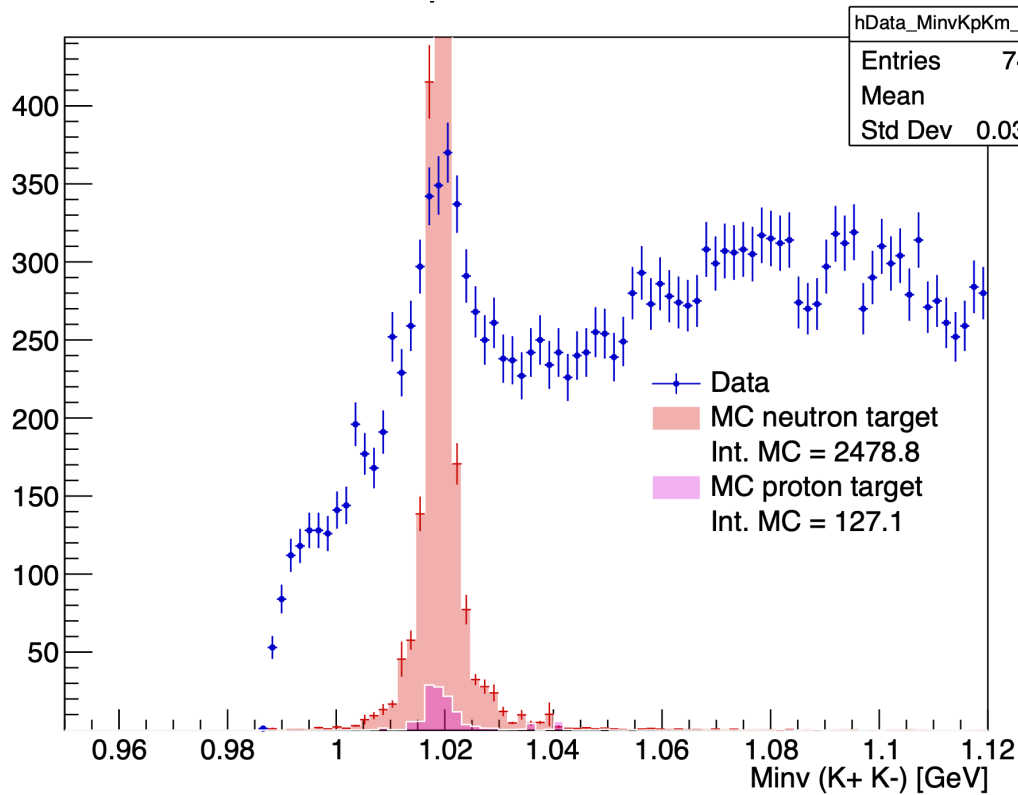
Good event with the angle between the missing nucleon and the neutron $< 5^\circ$



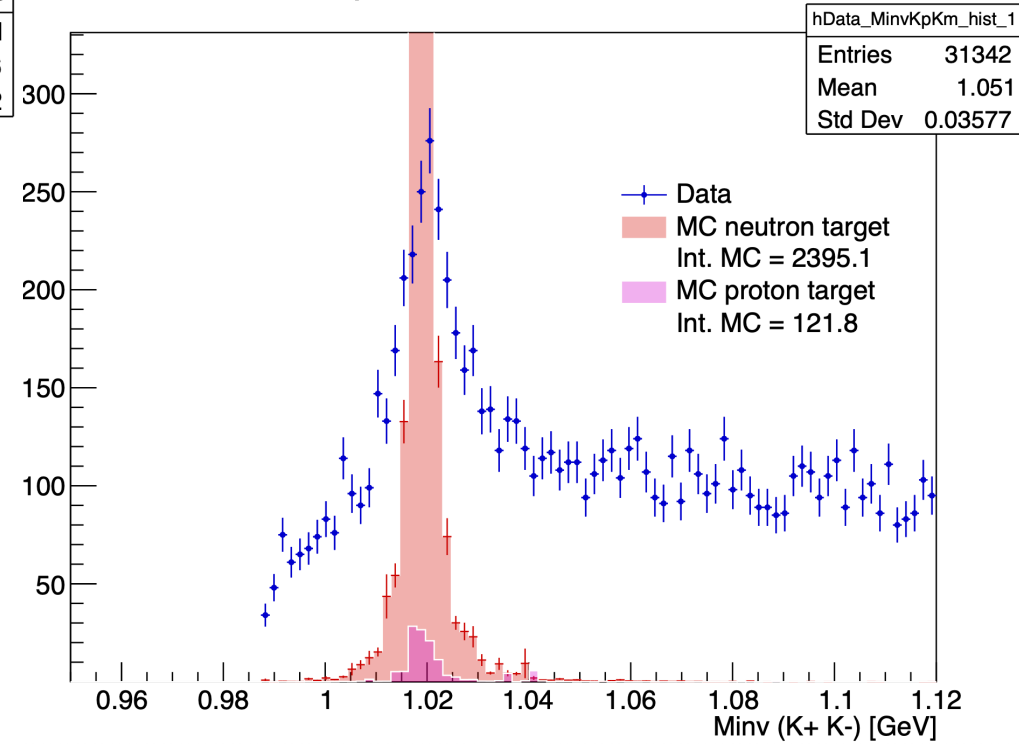
There is still a contamination when the **missing_nucleon** is a **non detected proton** and there is a **fake signal neutron** who pass the cut on the angle

→ Evaluate the contamination with simulation on **proton target** and apply the cuts of **neutron analysis** and see how many events pass the cuts

Influence of cuts on the invariant mass $K^+ K^-$:

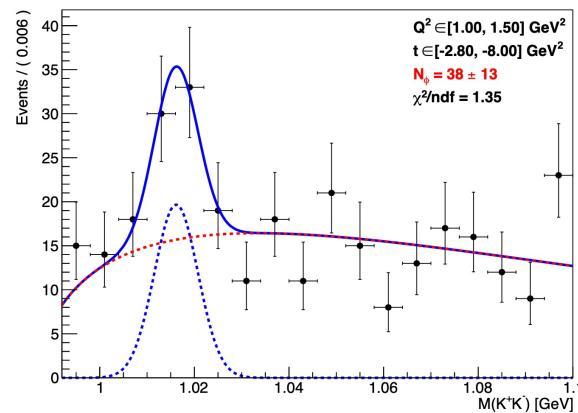
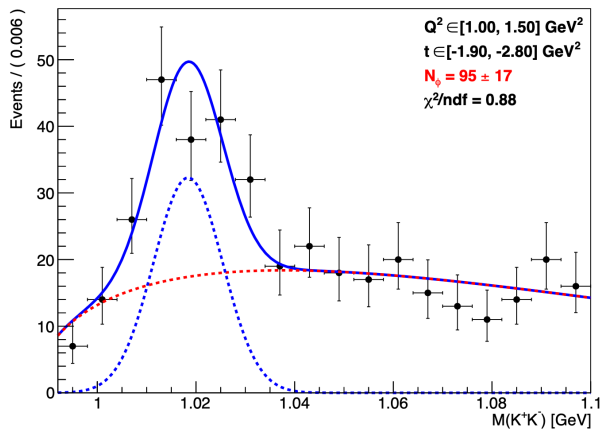
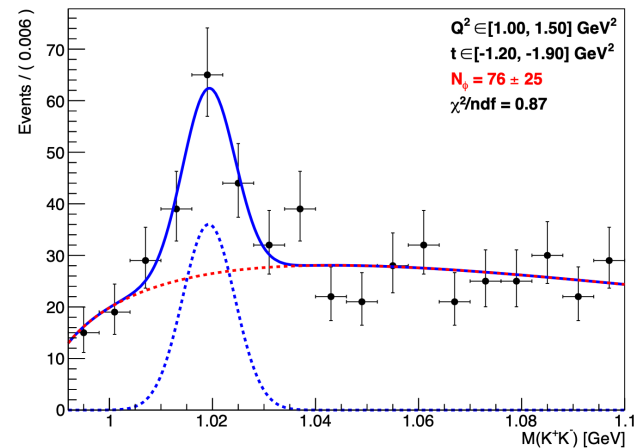
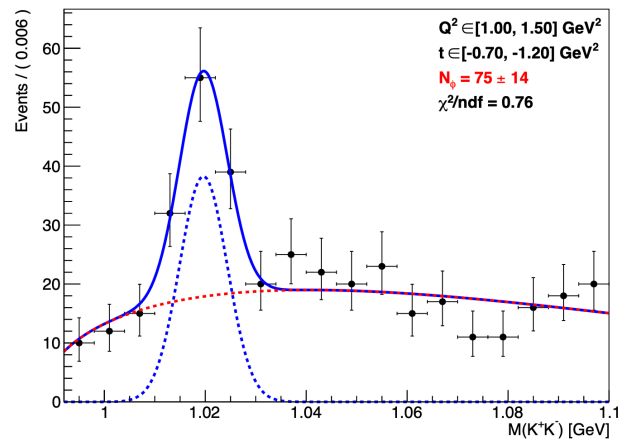
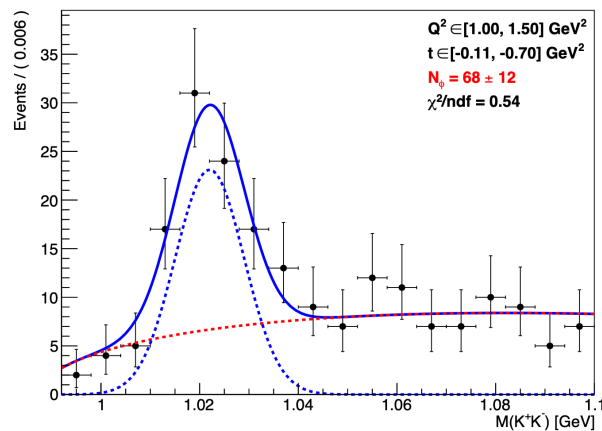


With angle cut

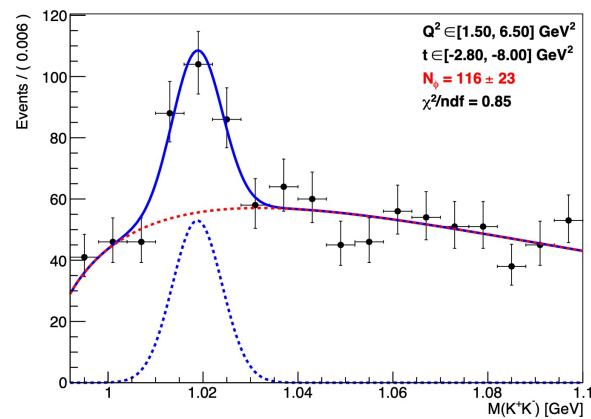
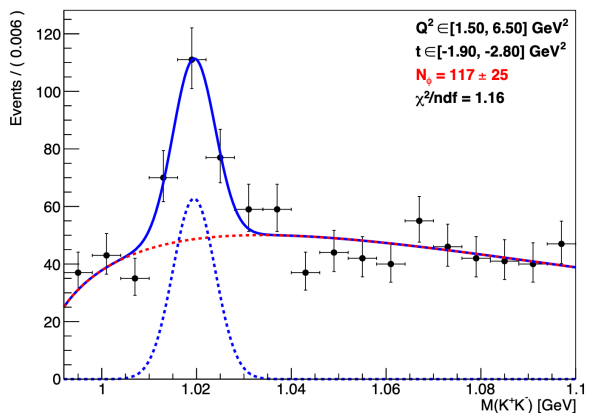
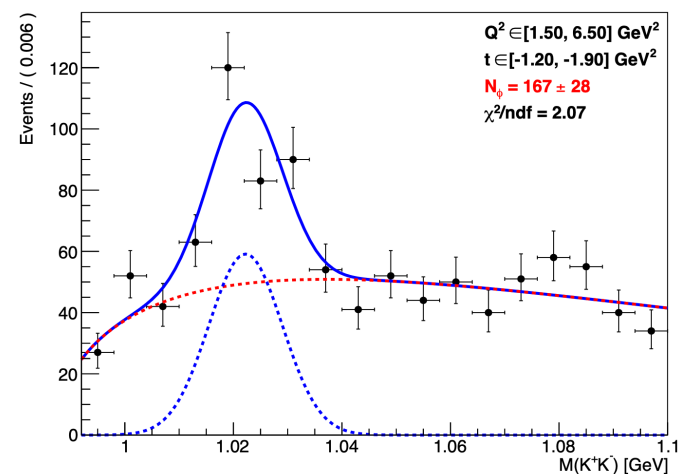
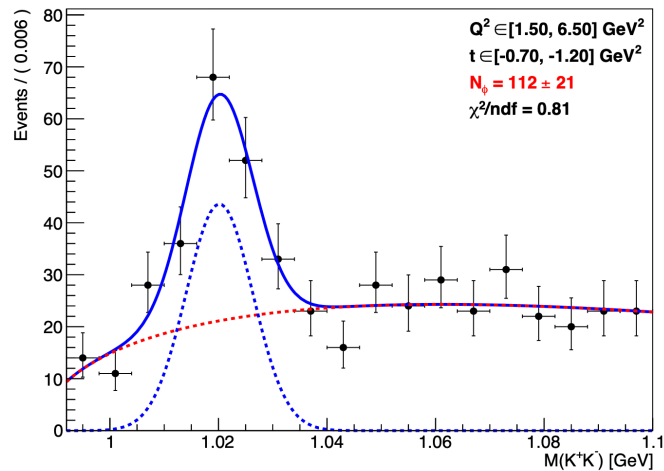
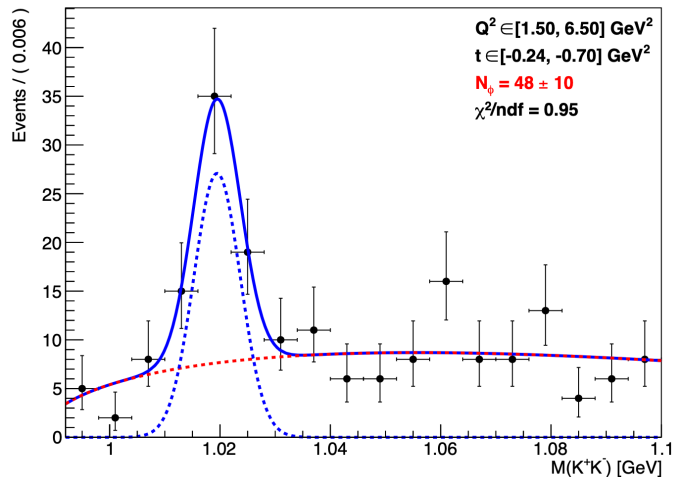


With angle cut and $K^+ K^-$ in FD

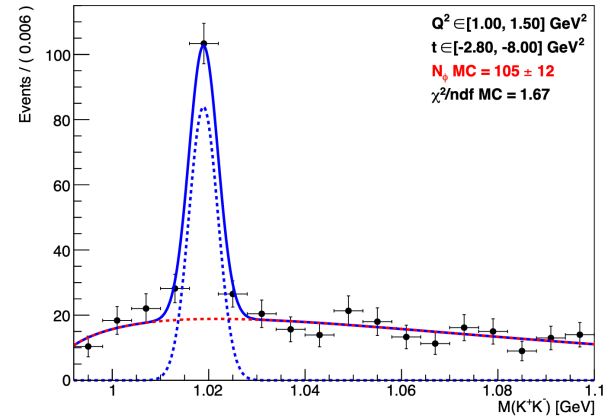
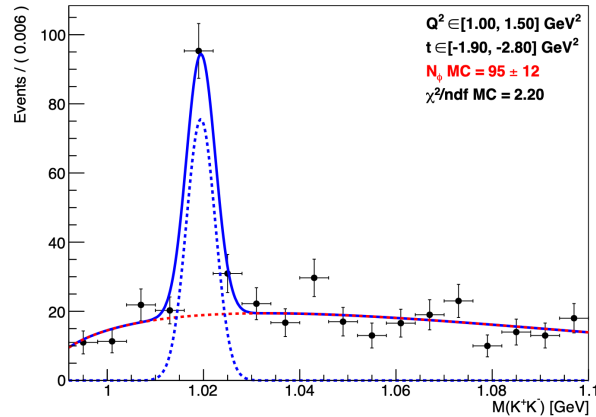
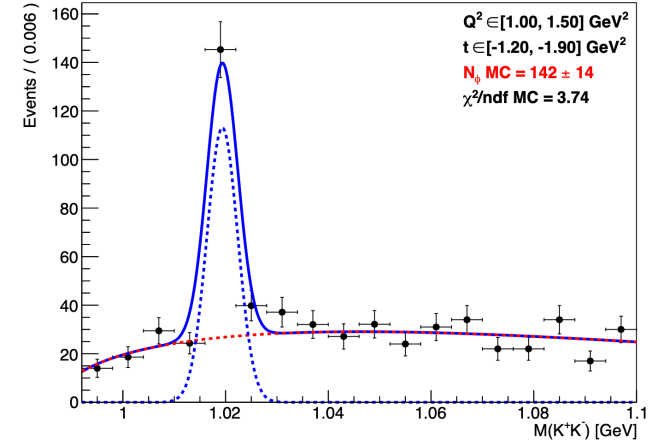
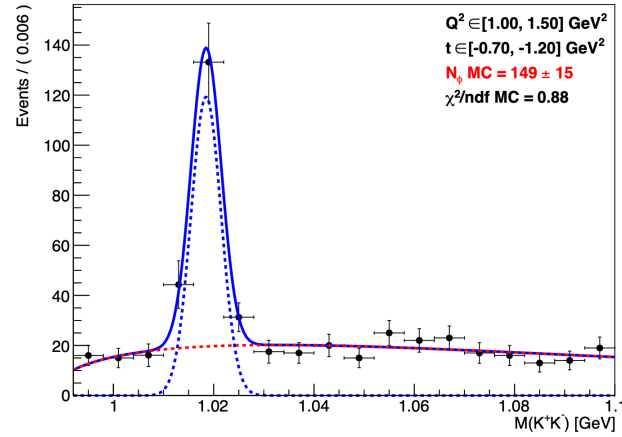
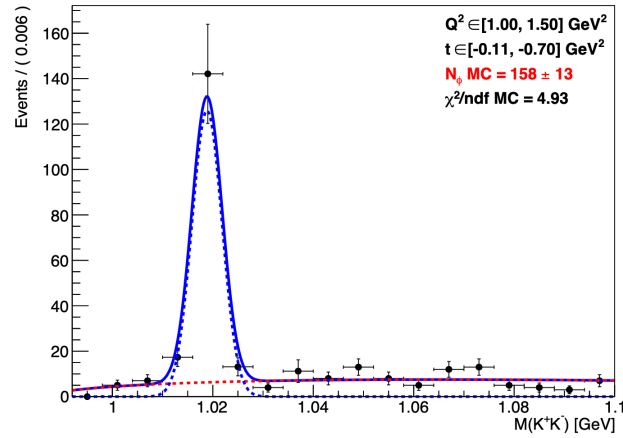
All fit data bin1 Q^2



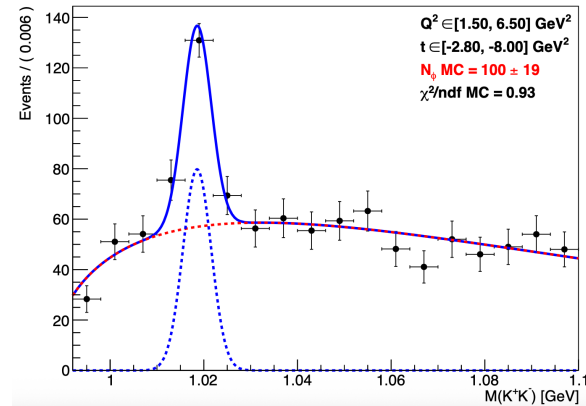
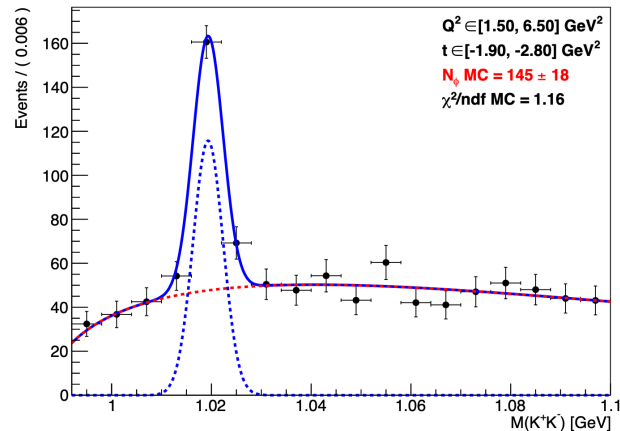
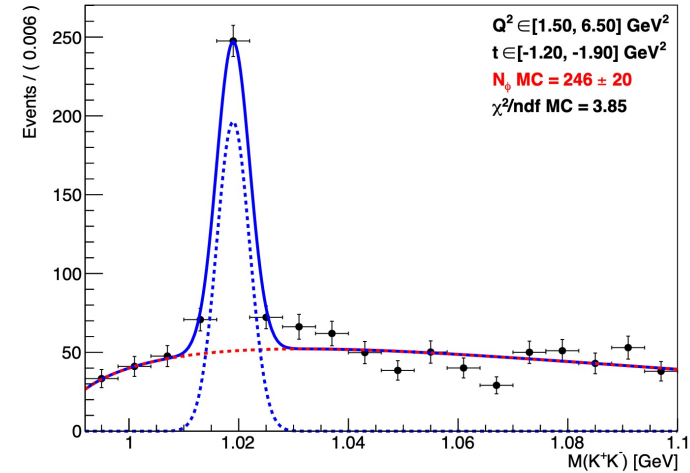
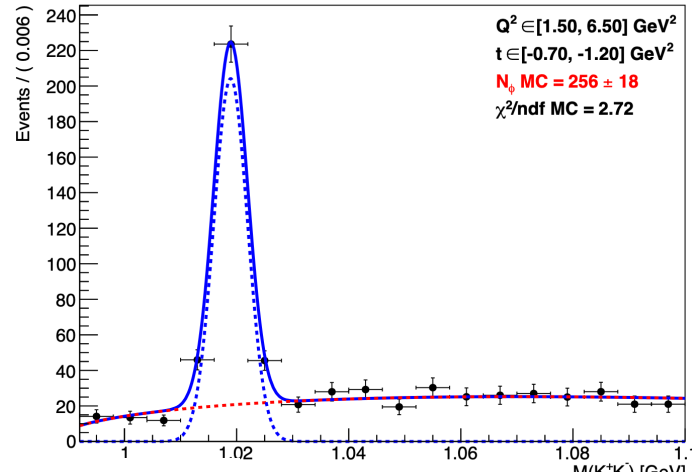
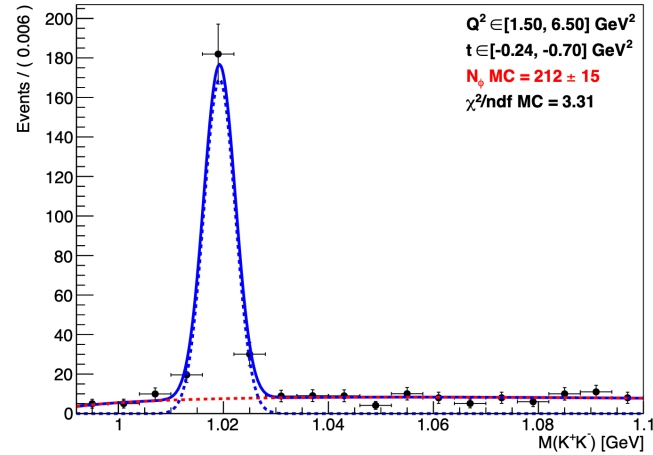
All fit data bin2 Q^2



All fit mc bin1 Q^2

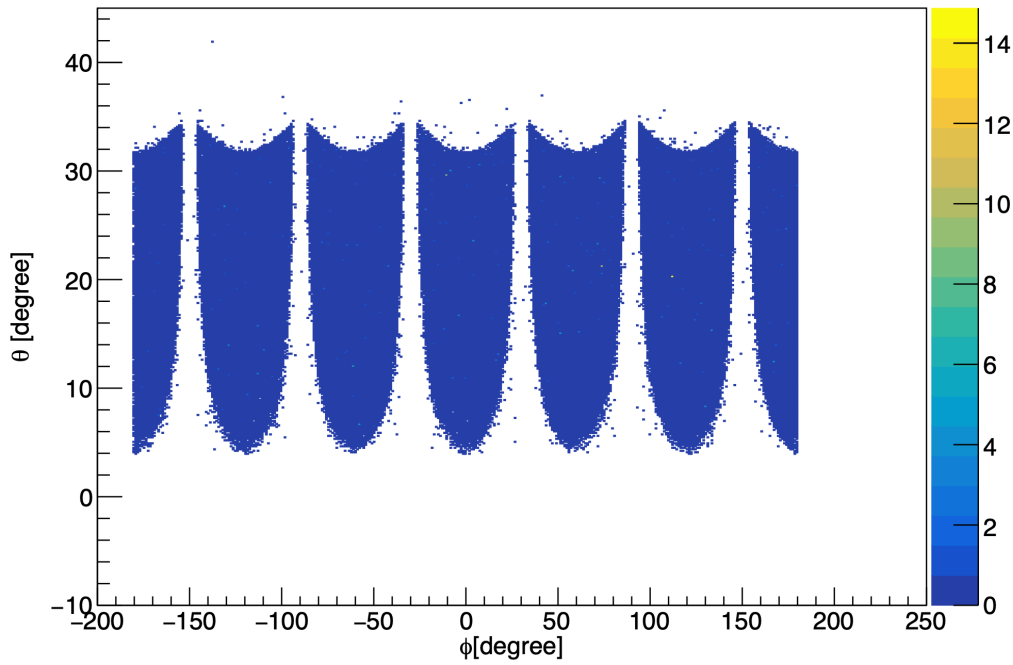


All fit mc bin2 Q²

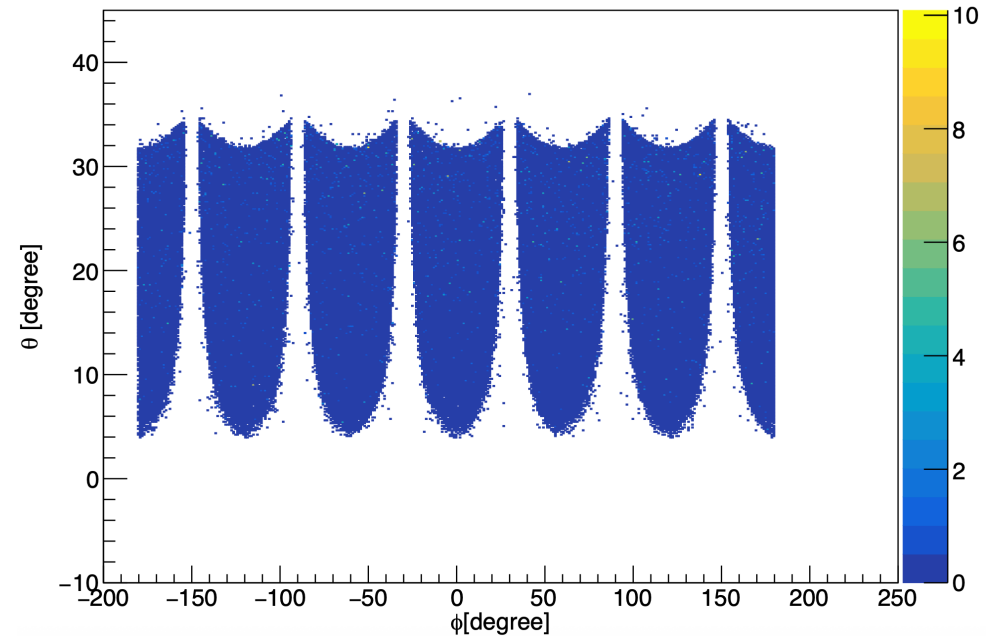


Neutron efficiency correction additional plots

theta vs phi neutron REC

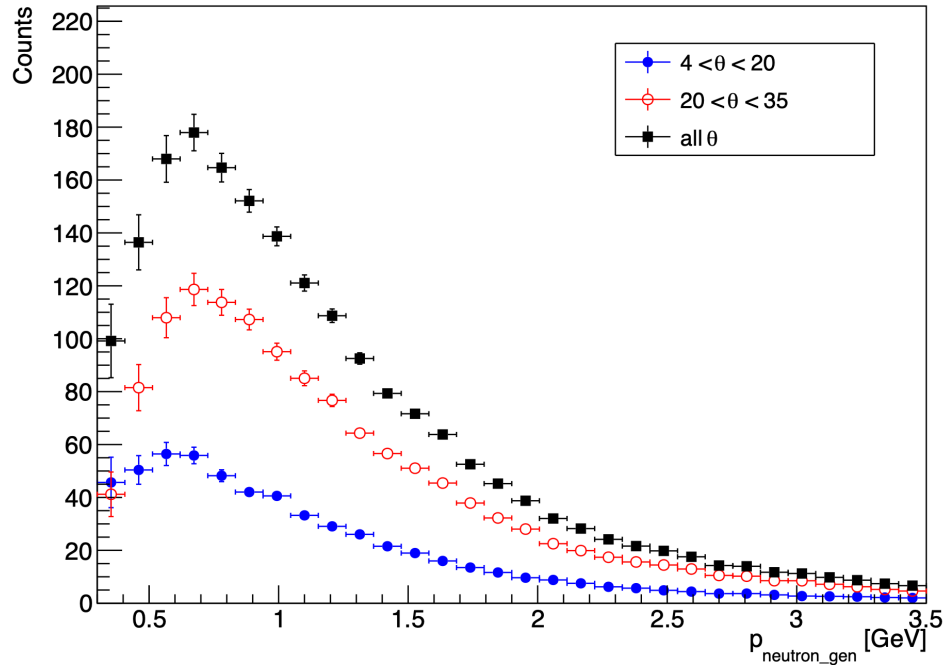


theta vs phi neutron GEN

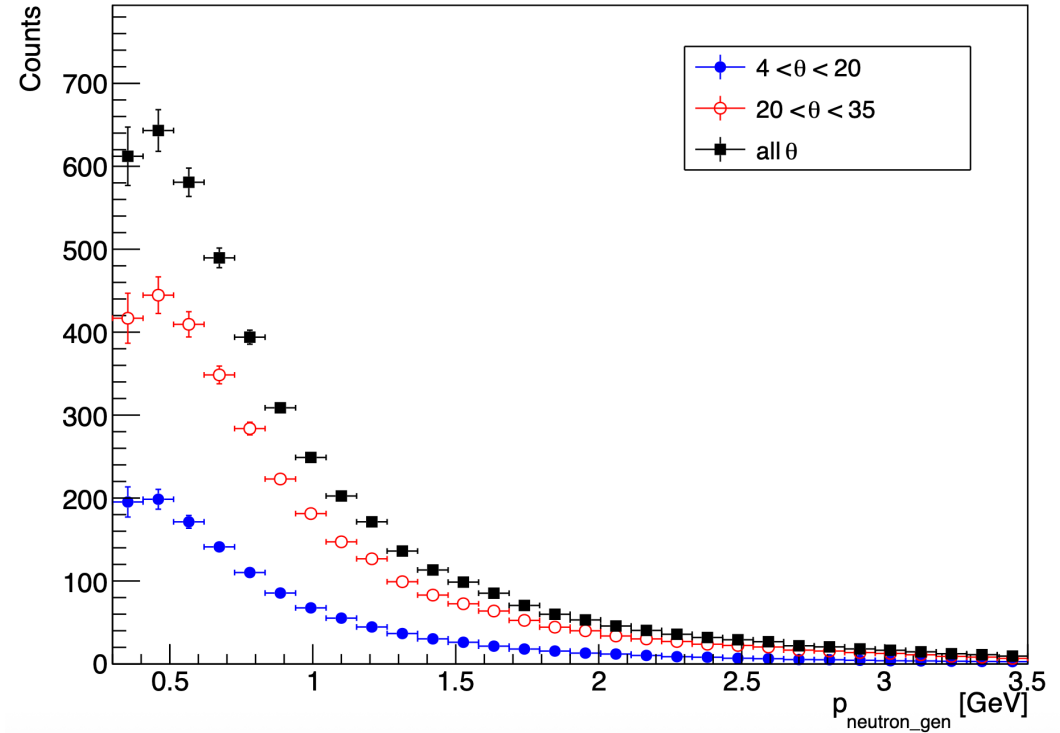


Neutron efficiency correction additional plots

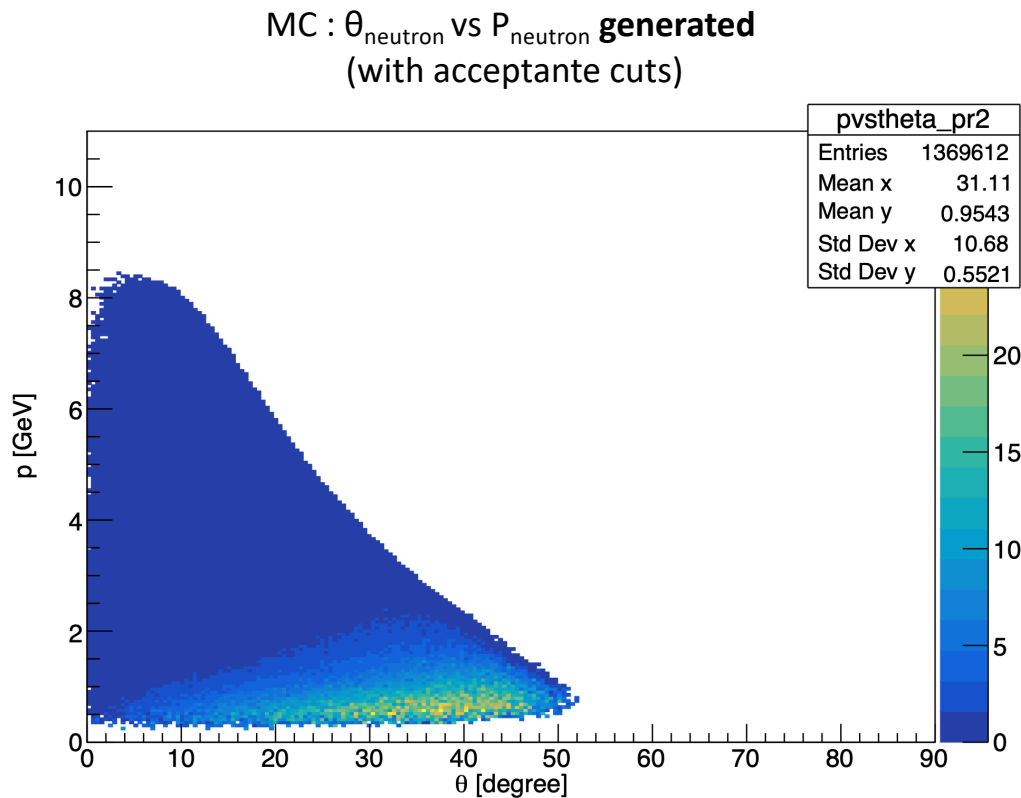
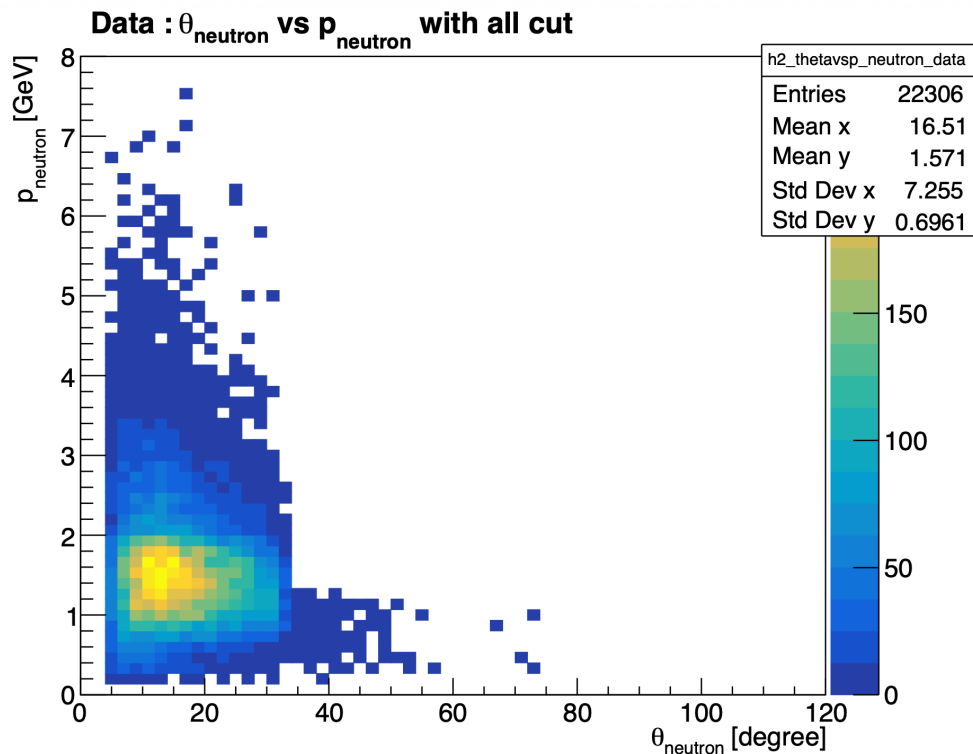
p neutron REC



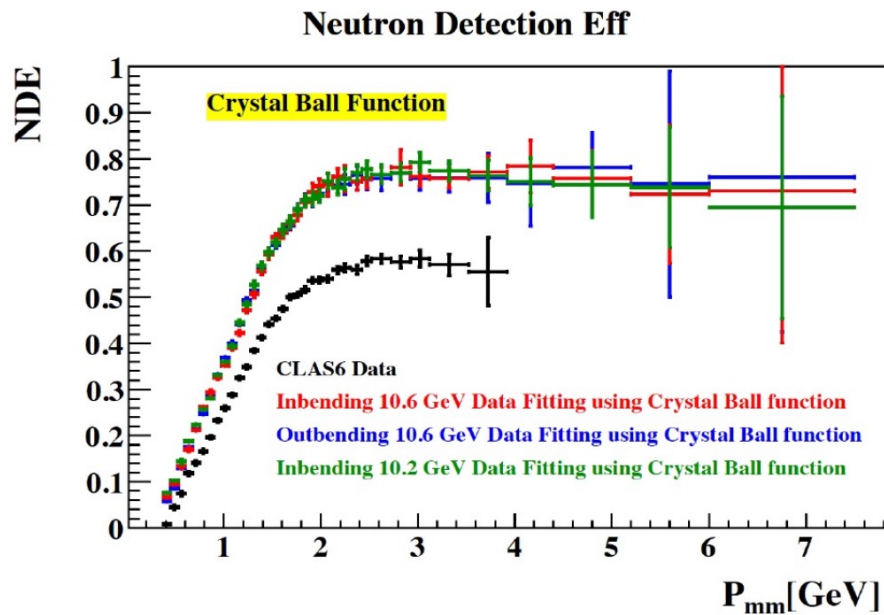
p neutron GEN



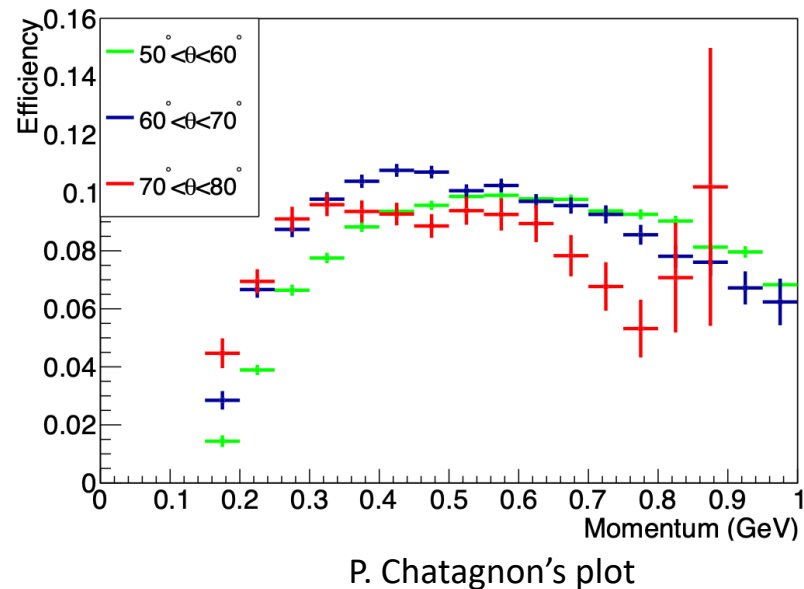
Neutron phasespace :



Neutron phasespace :



Neutron detection efficiency in the FD



Neutron detection efficiency in the CND